

Experiment No.	1
Experiment Title	Working with Python Packages – NumPy, SciPy, Scikit-Learn, Matplotlib
GitHub Repository	https://github.com/sangee-3010/Machine_Learning
Student Name	Sangeetha S K
Register Number	3122247001054
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	03/08/2026

1 Aim

To explore Python libraries such as NumPy, Pandas, SciPy, Scikit-learn and Matplotlib and apply them for data analysis, visualization, preprocessing and identification of suitable machine learning techniques for different public datasets.

2 Objectives

- Explore array manipulation, mathematical computing and data-handling operations available in NumPy, Pandas and SciPy.
- Understand how Scikit-learn structures a typical machine learning workflow, including preprocessing utilities such as `LabelEncoder`, `StandardScaler` and `MinMaxScaler`.
- Use Matplotlib to visualize dataset characteristics through histograms, box plots, bar charts, scatter plots and correlation heatmaps.
- Download and inspect five datasets from public repositories (Kaggle / UCI), and identify the type of machine learning task each one is suited to.
- Practice the initial stages of a machine learning workflow – loading data, exploratory data analysis, and preprocessing (missing value handling, encoding, and scaling) – on each dataset.

3 Introduction

Machine learning involves several steps such as data collection, preprocessing, visualization, feature selection and model building. Python provides various libraries that simplify these tasks. NumPy is used for numerical computations and array operations, Pandas helps in handling and analysing datasets using DataFrames, SciPy provides additional scientific and statistical functions, Scikit-learn contains preprocessing methods and machine learning algorithms, and Matplotlib is used for data visualization.

In this experiment, different publicly available datasets such as Loan Amount Prediction, Diabetes Prediction, Iris, Email Spam Classification and MNIST are analysed. The datasets are explored using Python libraries to understand their structure, identify the suitable machine learning task, perform preprocessing operations and study the characteristics of different types of data.

4 Python Packages Used

Table 1: Python Libraries Used in the Notebook

Library	Purpose
<code>pandas</code>	Loading CSV files into DataFrames; inspecting shape, dtypes, missing values, and summary statistics
<code>numpy</code>	Selecting numerical/non-numerical columns and supporting numerical operations
<code>matplotlib.pyplot</code>	Generating histograms, box plots, bar charts, scatter plots and correlation heatmaps
<code>LabelEncoder</code> (<code>sklearn.preprocessing</code>)	Encoding categorical (object-typed) columns into numeric labels
<code>StandardScaler</code> (<code>sklearn.preprocessing</code>)	Standardizing numerical features to zero mean and unit variance
<code>MinMaxScaler</code> (<code>sklearn.preprocessing</code>)	Normalizing numerical features to the $[0, 1]$ range

5 Datasets Used

Dataset	Raw Shape	Duplicate Rows	Source File
Loan Amount Prediction	614×13	0	<code>loan.csv</code>
Predicting Diabetes	768×9	0	<code>diabetes.csv</code>
Iris Dataset	150×6	0	<code>Iris.csv</code>
Classification of Email Spam	5572×5	403	<code>spam.csv</code>
Handwritten Character Recognition (MNIST, train split)	60000×785	0	<code>mnist_train.csv</code>

6 Machine Learning Workflow

The notebook follows the same six-stage workflow across all five datasets:

1. **Loading the Dataset:** Each CSV file is read into a Pandas DataFrame using `pd.read_csv()`, followed by an initial inspection using `head()`, `tail()`, `shape`, `columns`, `dtypes`, `info()` and `describe(include='all')`.
2. **Exploratory Data Analysis and Visualization:** Histograms and box plots are generated for every numerical column, bar charts for every low-cardinality column (10 or fewer unique values), a scatter plot between the first two numerical features, and a correlation heatmap across all numerical columns.

3. **Data Preprocessing:** Missing values are imputed (mode for categorical columns, median for numerical columns), categorical (object-typed) columns are label-encoded, and outliers are counted per column using the $1.5 \times \text{IQR}$ rule.
4. **Feature Selection:** Discussed conceptually for each dataset in Section 9, since this introductory experiment focuses on library exploration and EDA rather than fitting a specific estimator.
5. **Data Splitting:** Discussed conceptually in Section 10, in preparation for the model-training experiments that follow this one in the laboratory sequence.
6. **Performance Evaluation:** Discussed conceptually in Section 12, tied to the ML task type identified for each dataset.

Every processed dataset is finally standardized with `StandardScaler` and separately normalized with `MinMaxScaler`, so that both scaling conventions are available for whichever downstream algorithm is eventually chosen.

7 Exploratory Data Analysis

For each dataset, the notebook produces a histogram and a box plot for every numerical column (after label-encoding any categorical columns), a bar chart for every column with 15 or fewer unique values, one scatter plot between the first two numerical columns, and one correlation heatmap. All 111 figures generated by the notebook are reproduced below, grouped by dataset, each with a caption and a brief data-driven inference.

Loan Amount Prediction Dataset

The raw Loan Amount Prediction Dataset contains 614 rows and 13 columns, with 0 duplicate row(s) detected via `df.duplicated().sum()`.

Table 3: Missing Values – Loan Amount Prediction Dataset

Column	Missing Values	Percentage (%)
Gender	13	2.12
Married	3	0.49
Dependents	15	2.44
Self_Employed	32	5.21
LoanAmount	22	3.58
Loan_Amount_Term	14	2.28
Credit_History	50	8.14

Missing values were imputed using the column mode for categorical fields and the column median for numerical fields, after which no missing values remained in the dataset.

Histograms – Loan Amount Prediction Dataset

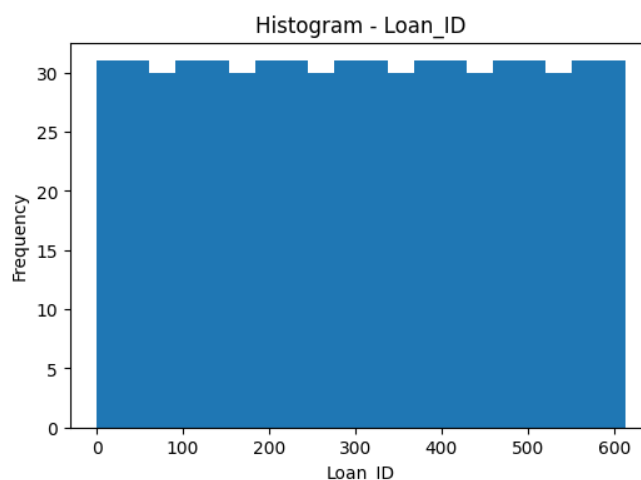


Figure 1: Histogram of Loan_ID – Loan Amount Prediction Dataset

Inference: Loan_ID a unique alphanumeric identifier assigned to every applicant, so it carries no real distributional meaning once label-encoded. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

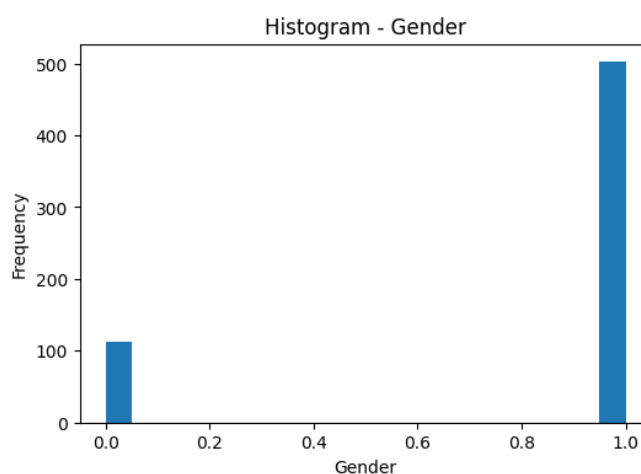


Figure 2: Histogram of Gender – Loan Amount Prediction Dataset

Inference: Gender encoded as a binary flag (Male/Female); of the 601 non-missing entries, 489 applicants were recorded as Male. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

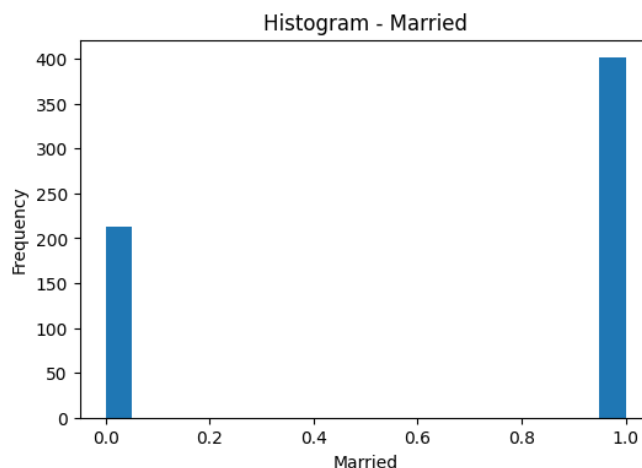


Figure 3: Histogram of Married – Loan Amount Prediction Dataset

Inference: Married encoded as a binary flag; 398 of 611 non-missing applicants were Married. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

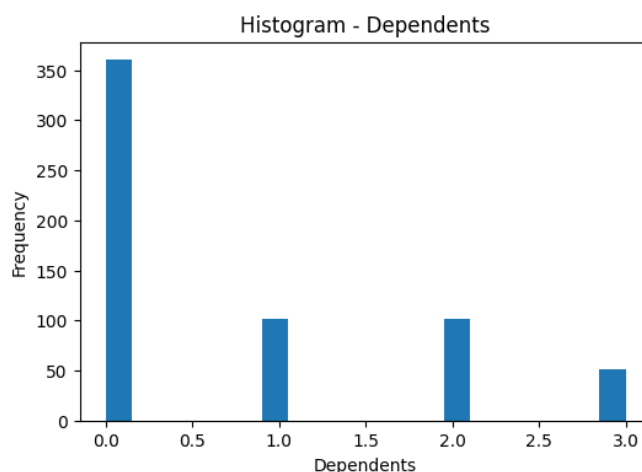


Figure 4: Histogram of Dependents – Loan Amount Prediction Dataset

Inference: Dependents encoded from the four original categories (0, 1, 2, 3+); '0' dependents was the most frequent category with 345 occurrences. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

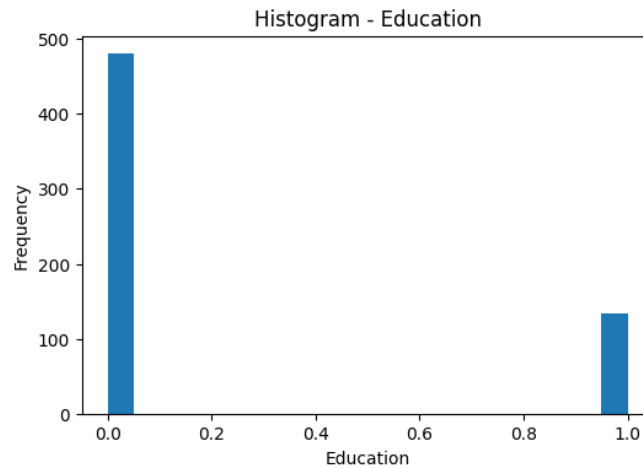


Figure 5: Histogram of Education – Loan Amount Prediction Dataset

Inference: Education encoded as Graduate/Not Graduate; 480 of 614 applicants were Graduates. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

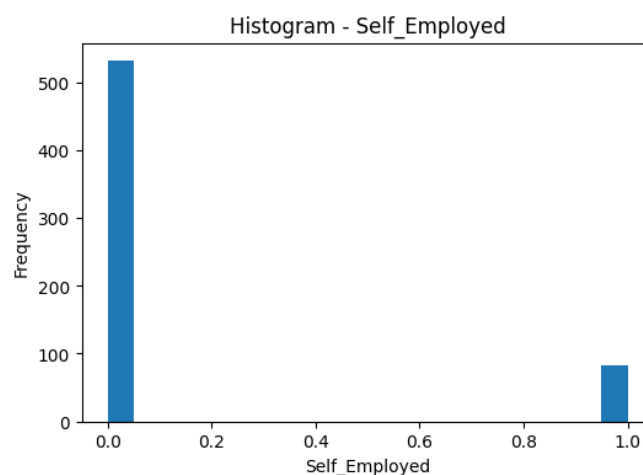


Figure 6: Histogram of Self_Employed – Loan Amount Prediction Dataset

Inference: Self_Employed encoded as a binary flag; 500 of 582 non-missing applicants were not self-employed. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

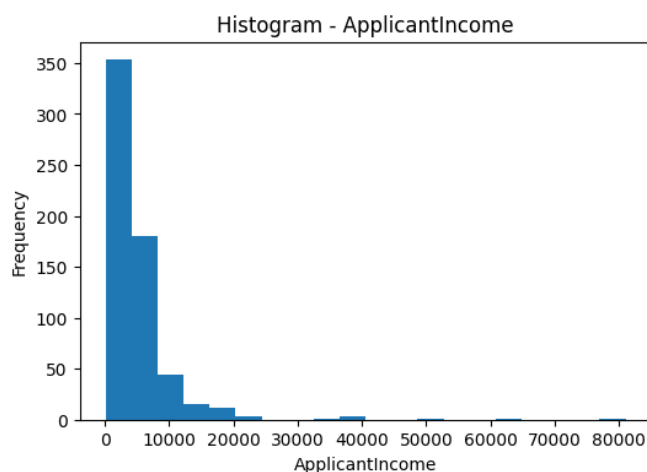


Figure 7: Histogram of ApplicantIncome – Loan Amount Prediction Dataset

Inference: ApplicantIncome ranges from 150 to 81000 with a mean of 5403.46 and a standard deviation of 6109.04, indicating a strongly right-skewed income distribution. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

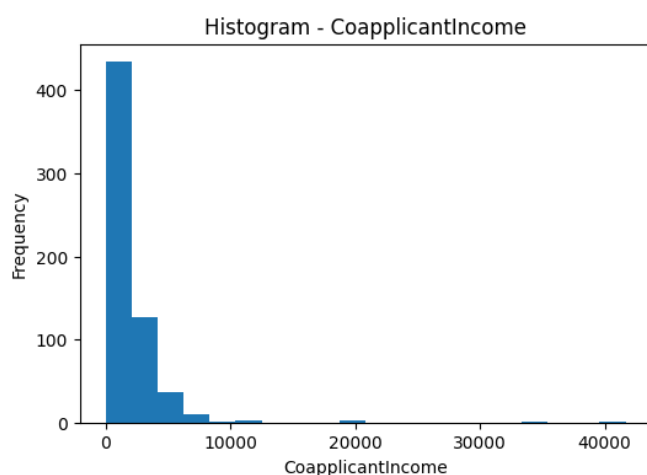


Figure 8: Histogram of CoapplicantIncome – Loan Amount Prediction Dataset

Inference: CoapplicantIncome ranges from 0 to 41667 with a mean of 1621.25 and a standard deviation of 2926.25; a large share of applicants report zero co-applicant income. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

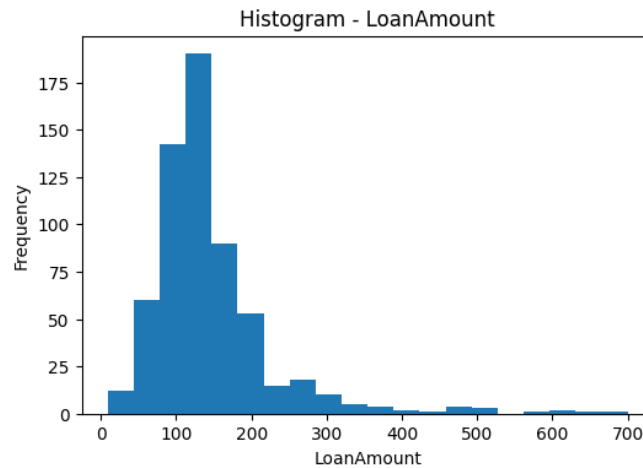


Figure 9: Histogram of LoanAmount – Loan Amount Prediction Dataset

Inference: LoanAmount ranges from 9 to 700 (in thousands) with a mean of 146.41 and a standard deviation of 85.59. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

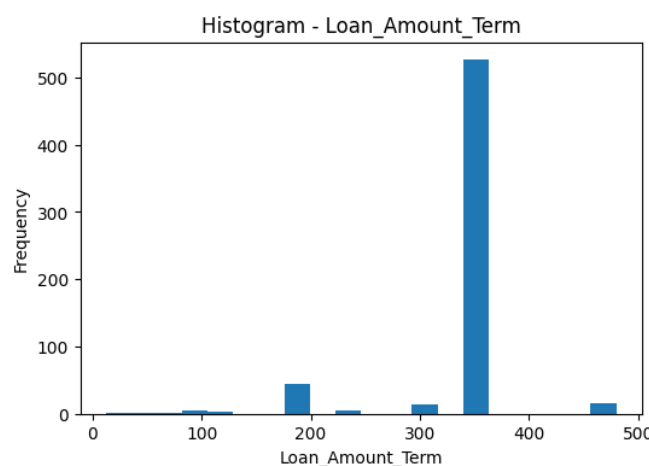


Figure 10: Histogram of Loan_Amount_Term – Loan Amount Prediction Dataset

Inference: Loan_Amount_Term is dominated by a term of 360 months, giving a mean of 342.0 and a comparatively small standard deviation of 65.12. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

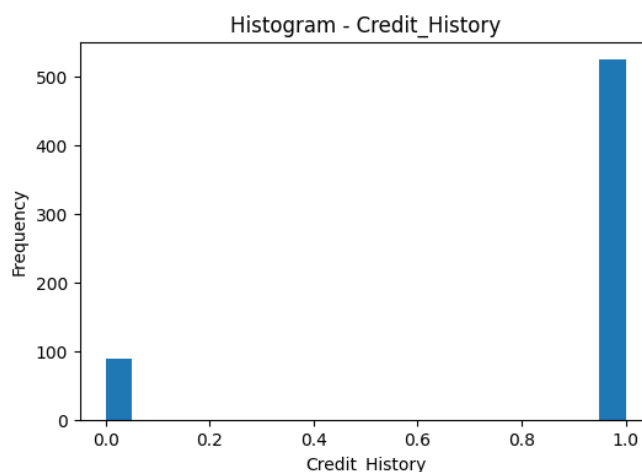


Figure 11: Histogram of Credit_History – Loan Amount Prediction Dataset

Inference: Credit_History is a near-binary flag where a value of 1 (favourable credit history) accounts for the vast majority of records, giving a mean of 0.842. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

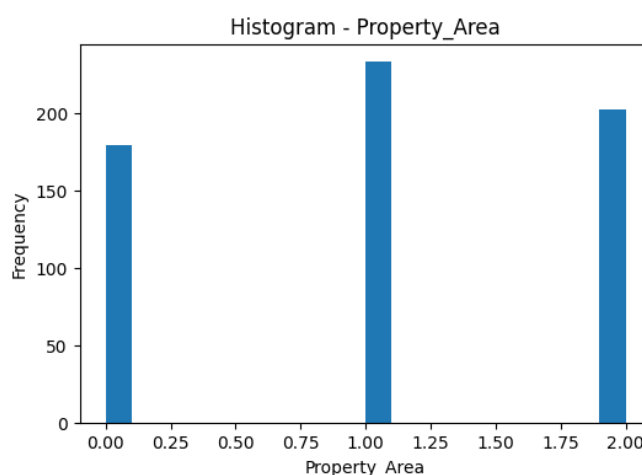


Figure 12: Histogram of Property_Area – Loan Amount Prediction Dataset

Inference: Property_Area encoded from three categories; Semiurban is the most common with 233 occurrences. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

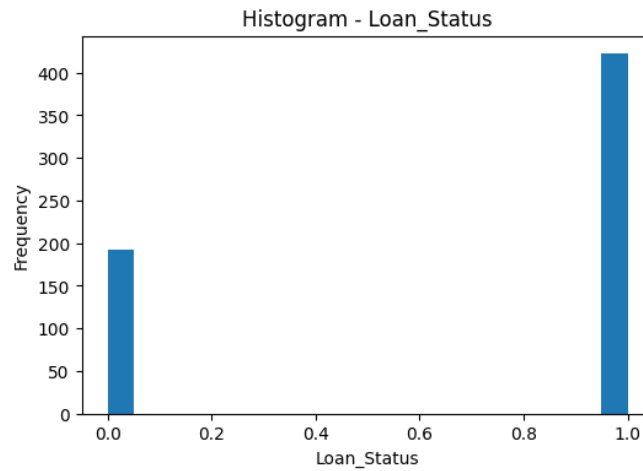


Figure 13: Histogram of Loan_Status – Loan Amount Prediction Dataset

Inference: Loan_Status the encoded target label; 422 of 614 applicants were approved (Y). Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

Box Plots – Loan Amount Prediction Dataset

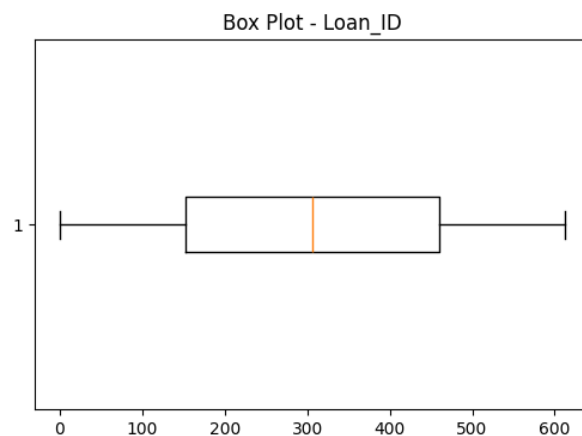


Figure 14: Box Plot of Loan_ID – Loan Amount Prediction Dataset

Inference: No values in Loan_ID were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

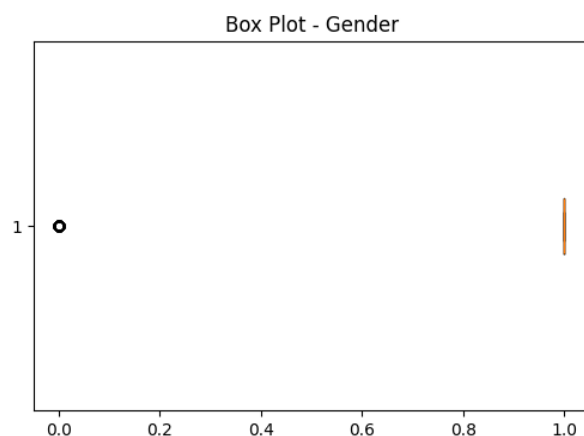


Figure 15: Box Plot of Gender – Loan Amount Prediction Dataset

Inference: The IQR-based outlier check flagged 112 value(s) in Gender lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Gender that standardization or robust scaling would need to account for.

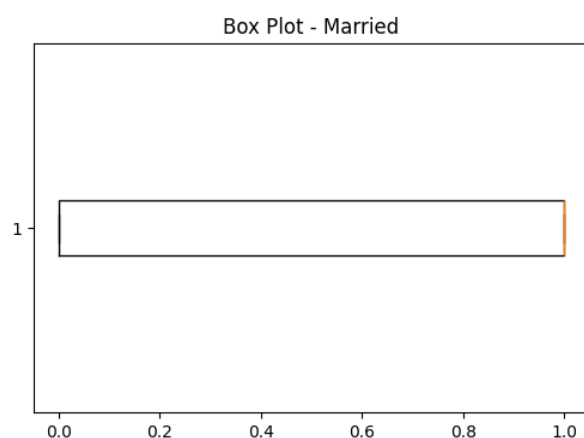


Figure 16: Box Plot of Married – Loan Amount Prediction Dataset

Inference: No values in Married were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

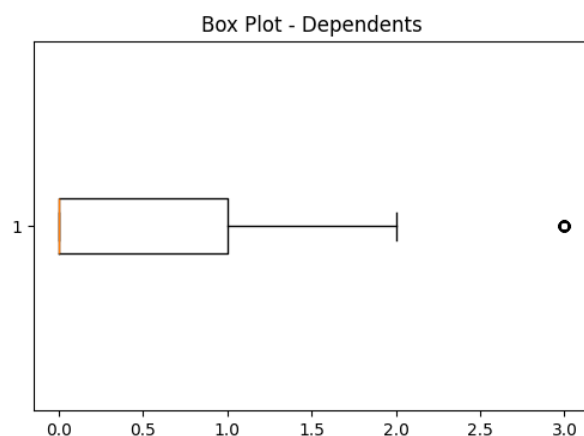


Figure 17: Box Plot of Dependents – Loan Amount Prediction Dataset

Inference: The IQR-based outlier check flagged 51 value(s) in Dependents lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Dependents that standardization or robust scaling would need to account for.

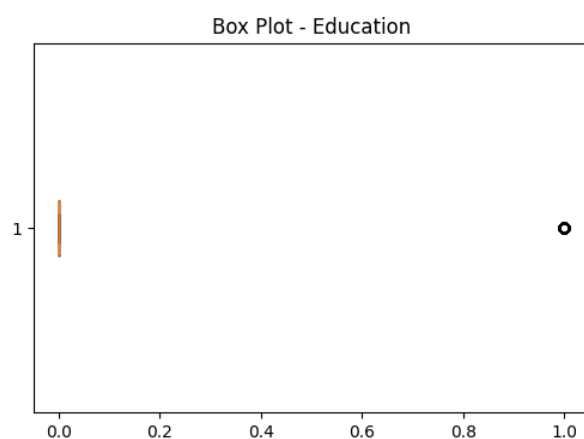


Figure 18: Box Plot of Education – Loan Amount Prediction Dataset

Inference: The IQR-based outlier check flagged 134 value(s) in Education lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Education that standardization or robust scaling would need to account for.

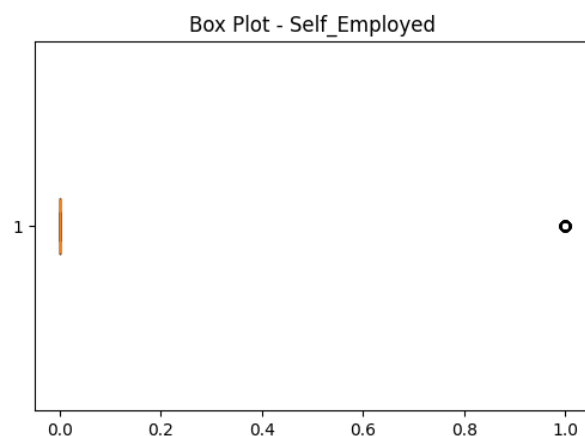


Figure 19: Box Plot of Self_Employed – Loan Amount Prediction Dataset

Inference: The IQR-based outlier check flagged 82 value(s) in Self_Employed lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Self_Employed that standardization or robust scaling would need to account for.

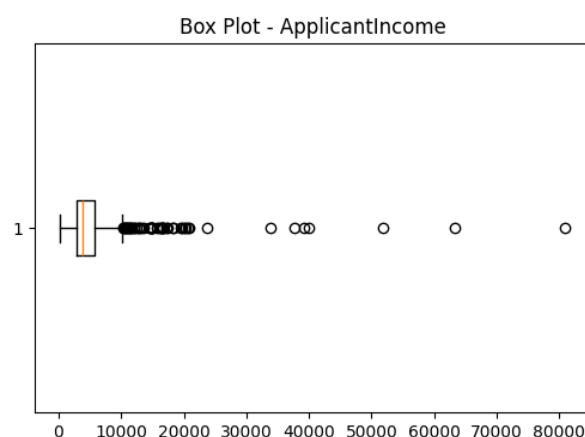


Figure 20: Box Plot of ApplicantIncome – Loan Amount Prediction Dataset

Inference: The IQR-based outlier check flagged 50 value(s) in ApplicantIncome lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of ApplicantIncome that standardization or robust scaling would need to account for.

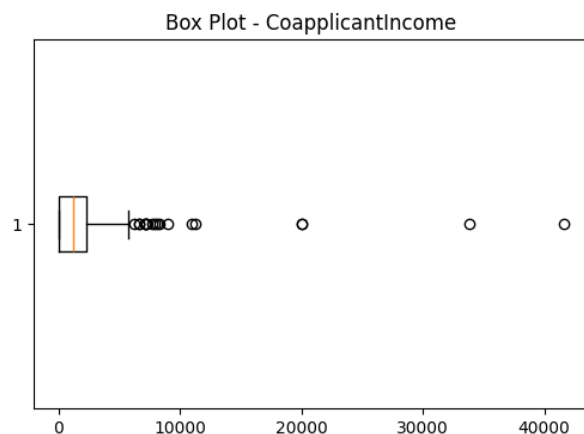


Figure 21: Box Plot of CoapplicantIncome – Loan Amount Prediction Dataset

Inference: The IQR-based outlier check flagged 18 value(s) in CoapplicantIncome lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of CoapplicantIncome that standardization or robust scaling would need to account for.

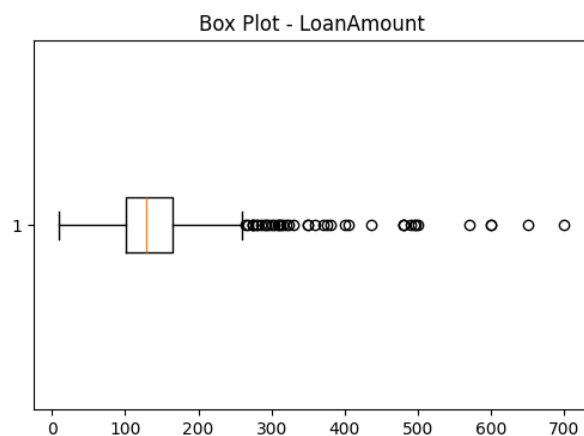


Figure 22: Box Plot of LoanAmount – Loan Amount Prediction Dataset

Inference: The IQR-based outlier check flagged 41 value(s) in LoanAmount lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of LoanAmount that standardization or robust scaling would need to account for.

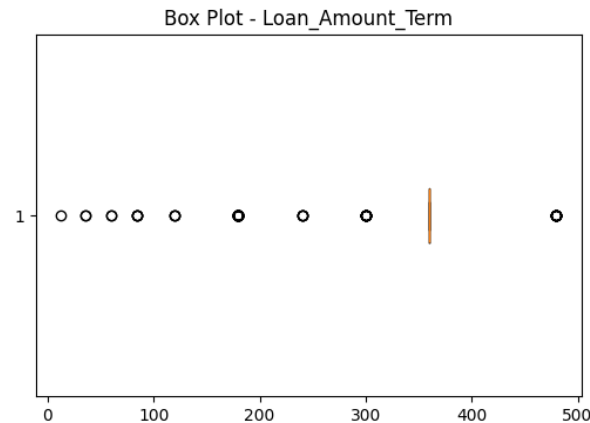


Figure 23: Box Plot of Loan_Amount_Term – Loan Amount Prediction Dataset

Inference: The IQR-based outlier check flagged 88 value(s) in Loan_Amount_Term lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Loan_Amount_Term that standardization or robust scaling would need to account for.



Figure 24: Box Plot of Credit_History – Loan Amount Prediction Dataset

Inference: The IQR-based outlier check flagged 89 value(s) in Credit_History lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Credit_History that standardization or robust scaling would need to account for.

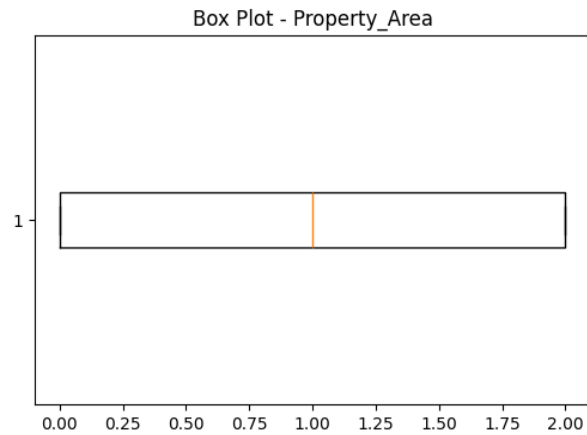


Figure 25: Box Plot of Property_Area – Loan Amount Prediction Dataset

Inference: No values in Property_Area were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

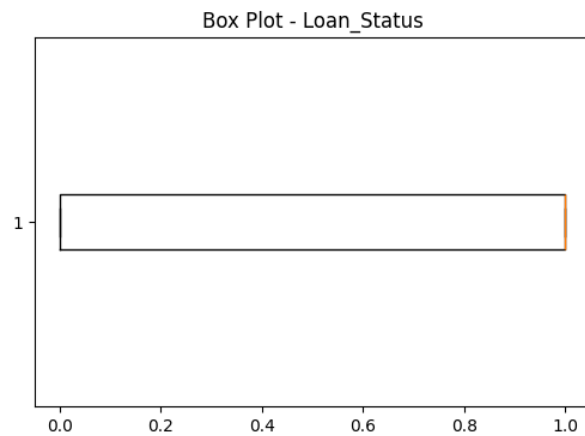


Figure 26: Box Plot of Loan_Status – Loan Amount Prediction Dataset

Inference: No values in Loan_Status were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

Bar Charts – Loan Amount Prediction Dataset

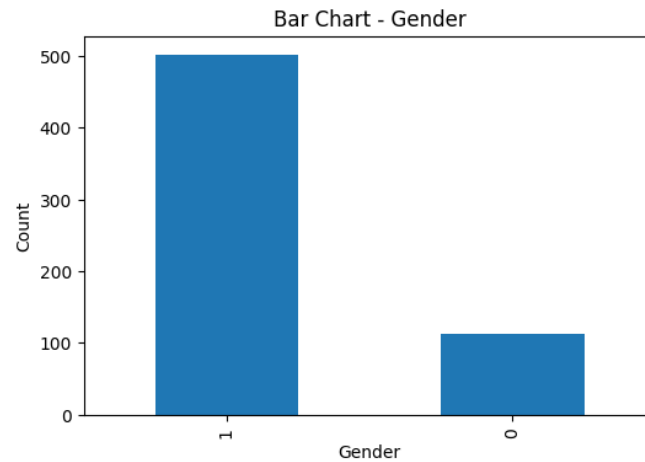


Figure 27: Bar Chart of Gender – Loan Amount Prediction Dataset

Inference: the applicant pool is heavily skewed toward Male applicants (489 of 601 recorded), with far fewer Female applicants.

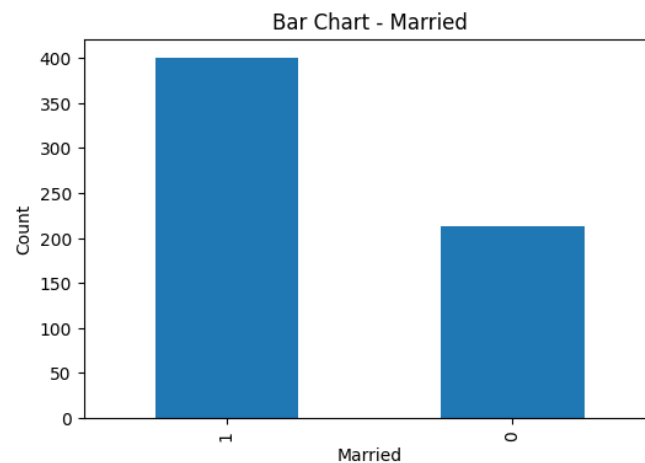


Figure 28: Bar Chart of Married – Loan Amount Prediction Dataset

Inference: Married applicants (398) outnumber unmarried applicants (213) by roughly a 2:1 ratio.

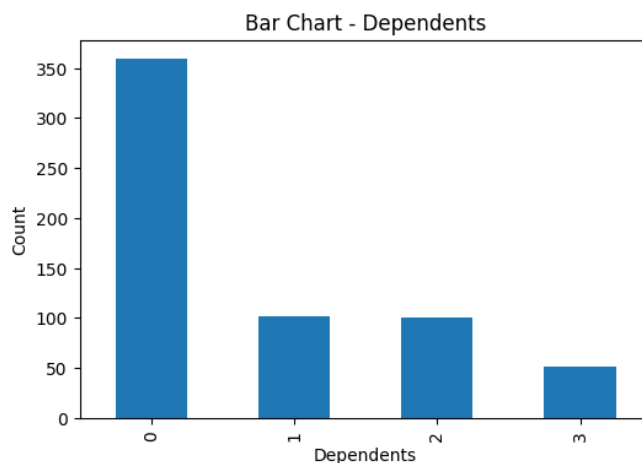


Figure 29: Bar Chart of Dependents – Loan Amount Prediction Dataset

Inference: applicants with zero dependents form the largest group (345), followed by a steep drop-off for 1, 2, and 3+ dependents.

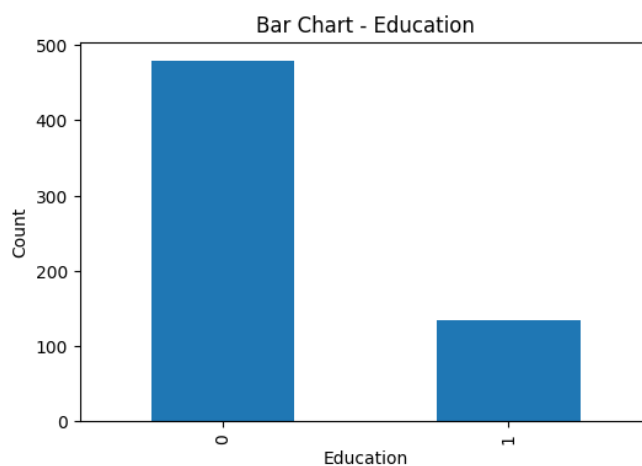


Figure 30: Bar Chart of Education – Loan Amount Prediction Dataset

Inference: Graduate applicants (480) considerably outnumber Not Graduate applicants (134).

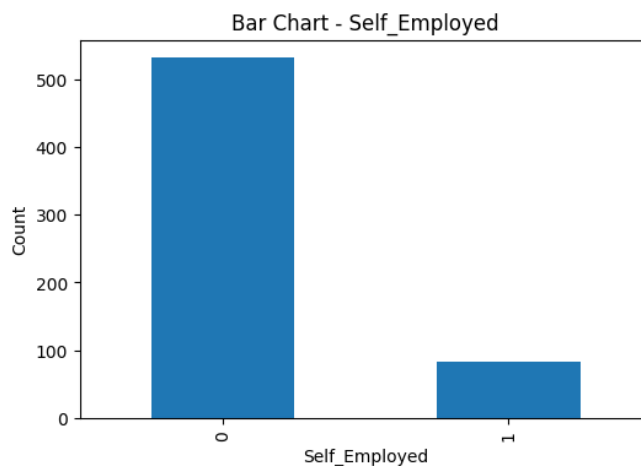


Figure 31: Bar Chart of Self_Employed – Loan Amount Prediction Dataset

Inference: the large majority of applicants (500) are not self-employed, with only 82 self-employed.

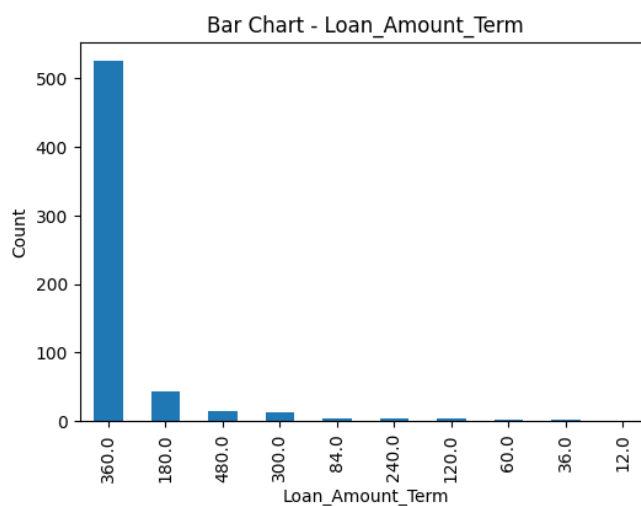


Figure 32: Bar Chart of Loan_Amount_Term – Loan Amount Prediction Dataset

Inference: a 360-month term dominates the dataset almost completely, with the remaining terms (e.g. 180, 480) appearing only sporadically.

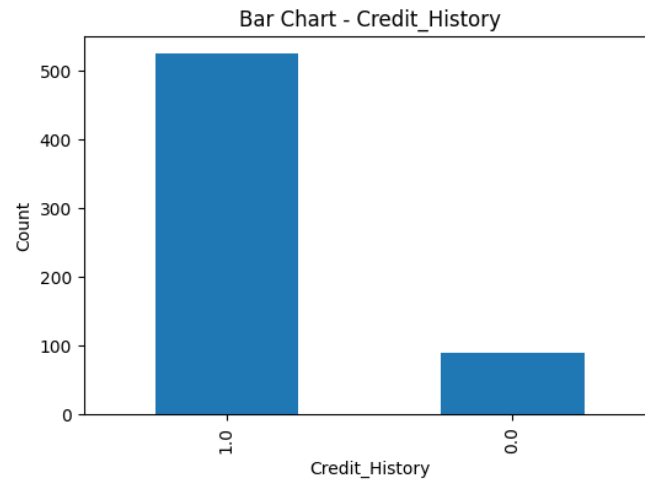


Figure 33: Bar Chart of Credit_History – Loan Amount Prediction Dataset

Inference: applicants with a favourable credit history (value 1) vastly outnumber those without, reflecting the 0.842 mean noted earlier.

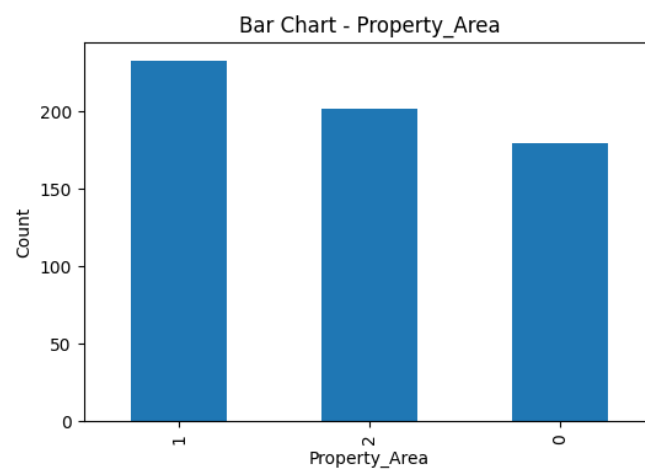


Figure 34: Bar Chart of Property_Area – Loan Amount Prediction Dataset

Inference: Semiurban properties (233) are slightly more common than Urban and Rural, which are fairly close to each other.

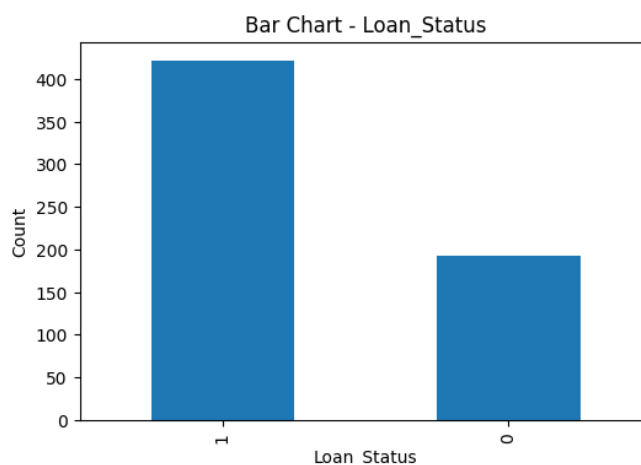


Figure 35: Bar Chart of Loan_Status – Loan Amount Prediction Dataset

Inference: approved loans (Y, 422) outnumber rejected loans (N, 192), so the target is moderately imbalanced.

Scatter Plot – Loan Amount Prediction Dataset

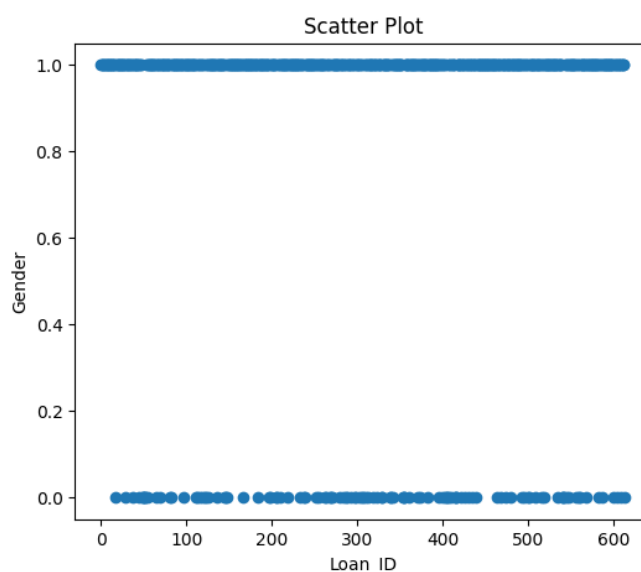


Figure 36: Scatter Plot of Loan_ID vs Gender – Loan Amount Prediction Dataset

Inference: Since the first two encoded numerical columns are Loan_ID (a unique row identifier) and Gender (a binary flag), the scatter plot shows two flat horizontal bands rather than a meaningful trend – it is a by-product of column ordering rather than a genuinely informative relationship, and ApplicantIncome vs LoanAmount would be a more meaningful pairing for this dataset.

Correlation Heatmap – Loan Amount Prediction Dataset

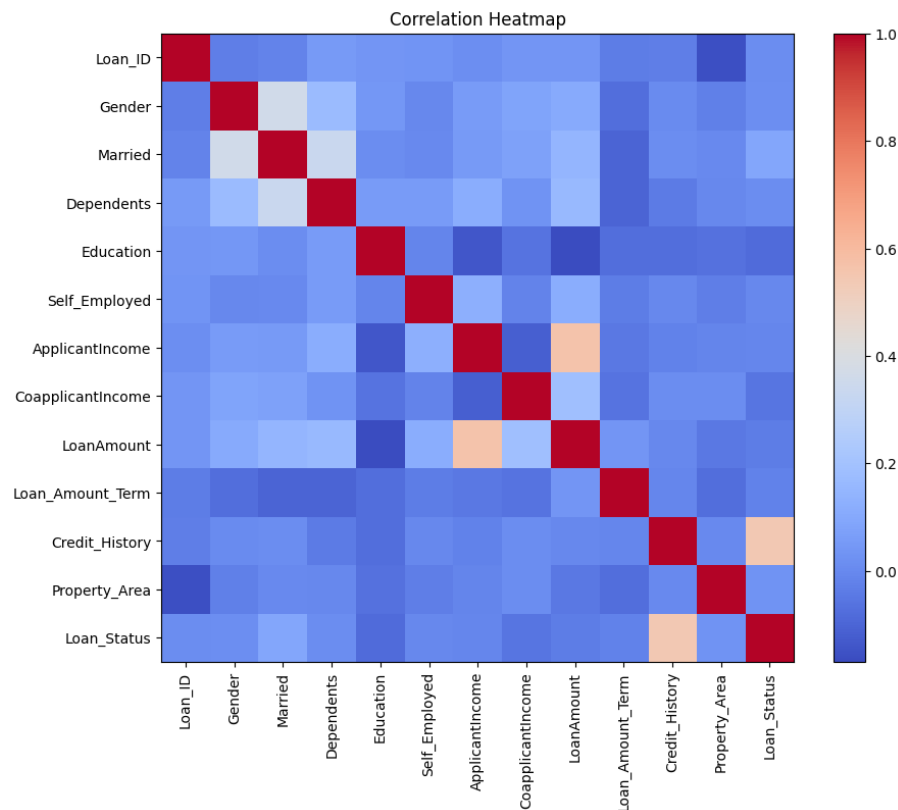


Figure 37: Correlation Heatmap – Loan Amount Prediction Dataset

Inference: The heatmap is dominated by the identifier column `Loan_ID`, which by construction shows negligible correlation with every other feature, while `ApplicantIncome` and `LoanAmount` show a mild positive correlation, consistent with larger incomes supporting larger loan requests. `Credit_History` stands out as the feature most associated with `Loan_Status`, reinforcing its known importance in loan approval decisions.

Pima Indians Diabetes Dataset

The raw Pima Indians Diabetes Dataset contains 768 rows and 9 columns, with 0 duplicate row(s) detected via `df.duplicated().sum()`. No missing values were detected in this dataset, so no imputation was required.

Histograms – Pima Indians Diabetes Dataset

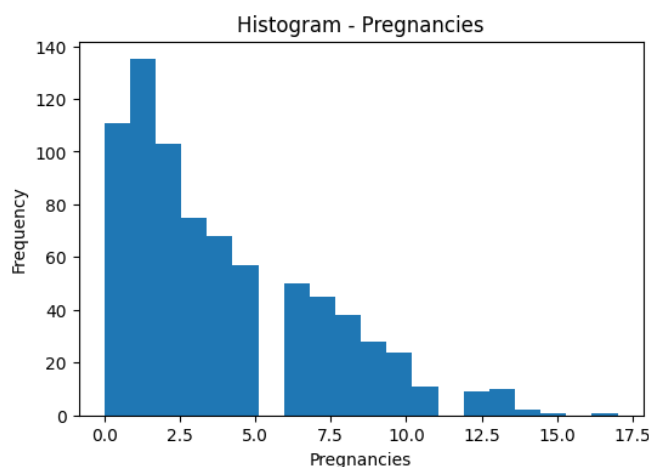


Figure 38: Histogram of Pregnancies – Pima Indians Diabetes Dataset

Inference: Pregnancies ranges from 0 to 17 with a mean of 3.85 and a standard deviation of 3.37, showing a right-skewed distribution typical of count data. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

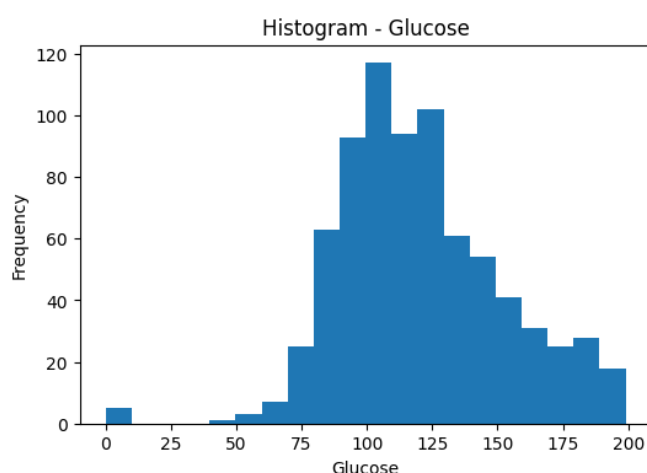


Figure 39: Histogram of Glucose – Pima Indians Diabetes Dataset

Inference: Glucose ranges from 0 to 199 with a mean of 120.89 and a standard deviation of 31.97; the presence of a minimum of 0 is almost certainly a placeholder for a missing clinical reading rather than a true value. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

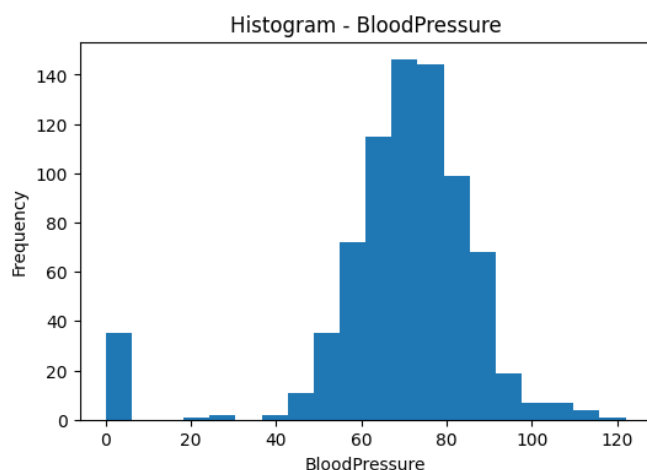


Figure 40: Histogram of BloodPressure – Pima Indians Diabetes Dataset

Inference: BloodPressure ranges from 0 to 122 with a mean of 69.11 and a standard deviation of 19.36, again with 0 values likely representing missing measurements. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

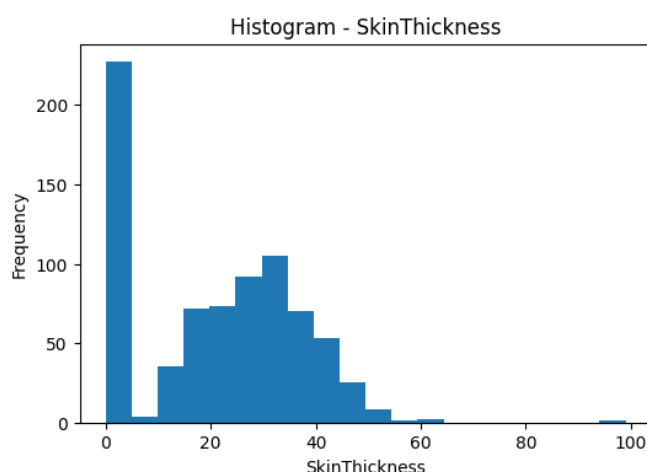


Figure 41: Histogram of SkinThickness – Pima Indians Diabetes Dataset

Inference: SkinThickness ranges from 0 to 99 with a mean of 20.54 and a standard deviation of 15.95, and shows a large spike at 0 that suggests many unrecorded measurements. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

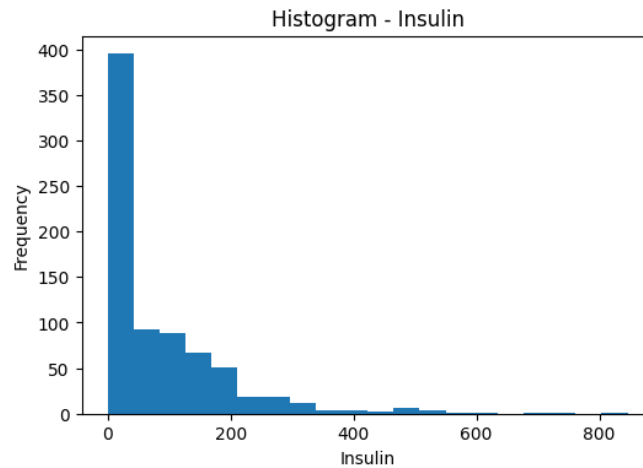


Figure 42: Histogram of Insulin – Pima Indians Diabetes Dataset

Inference: Insulin ranges from 0 to 846 with a mean of 79.80 and a very large standard deviation of 115.24, indicating heavy right skew and a substantial cluster of zero readings. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

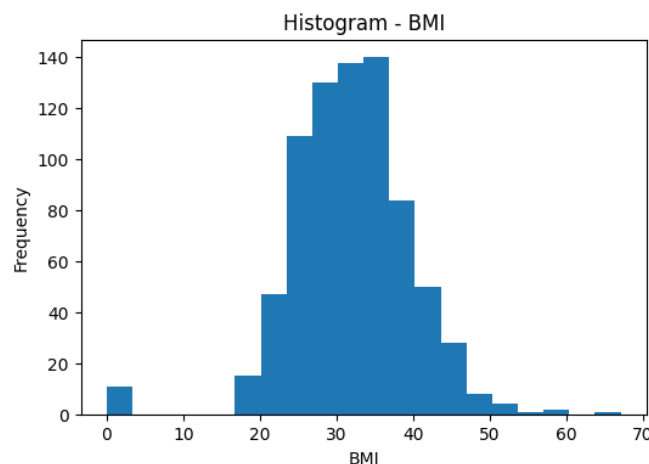


Figure 43: Histogram of BMI – Pima Indians Diabetes Dataset

Inference: BMI ranges from 0 to 67.1 with a mean of 31.99 and a standard deviation of 7.88, roughly centered but with a small number of implausible zero entries. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

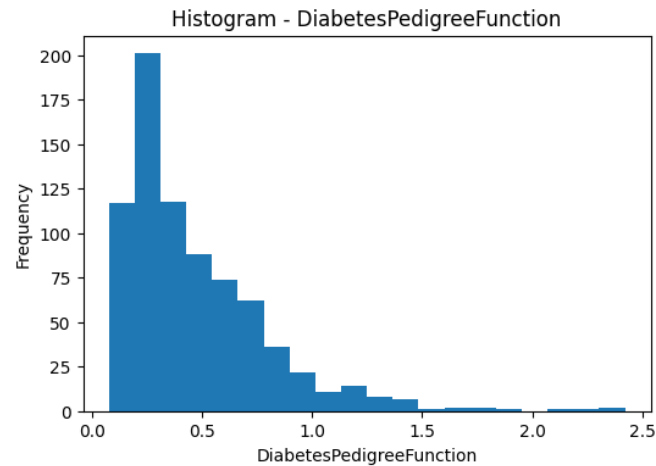


Figure 44: Histogram of DiabetesPedigreeFunction – Pima Indians Diabetes Dataset

Inference: DiabetesPedigreeFunction ranges from 0.078 to 2.42 with a mean of 0.472 and a standard deviation of 0.331, showing a strongly right-skewed distribution. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

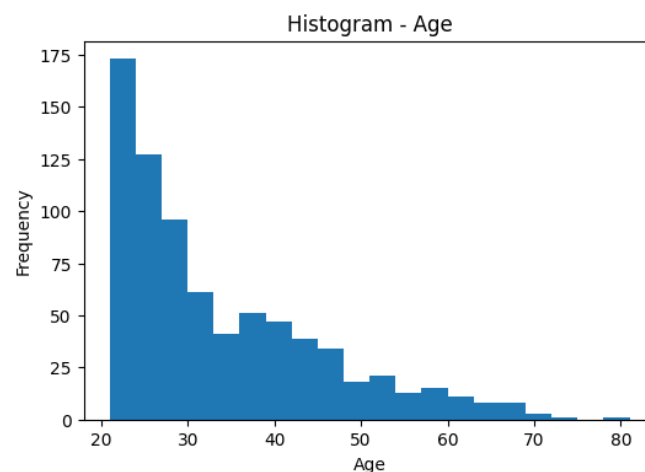


Figure 45: Histogram of Age – Pima Indians Diabetes Dataset

Inference: Age ranges from 21 to 81 with a mean of 33.24 and a standard deviation of 11.76, skewed toward younger applicants. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

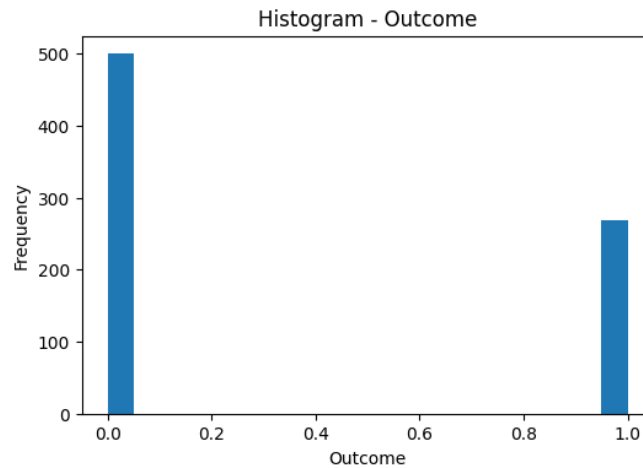


Figure 46: Histogram of Outcome – Pima Indians Diabetes Dataset

Inference: Outcome the binary diabetes diagnosis label; 268 of 768 patients (34.9%) were diagnosed positive. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

Box Plots – Pima Indians Diabetes Dataset

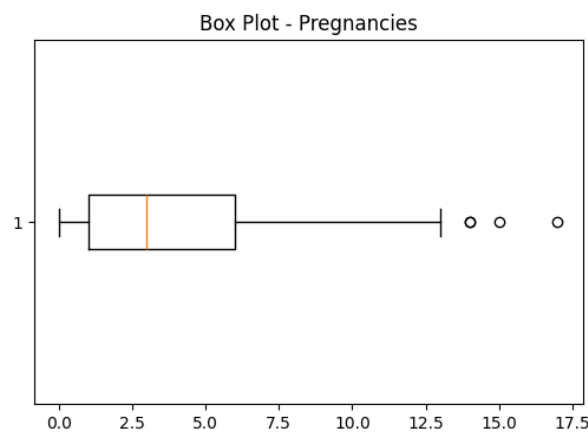


Figure 47: Box Plot of Pregnancies – Pima Indians Diabetes Dataset

Inference: The IQR-based outlier check flagged 4 value(s) in Pregnancies lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Pregnancies that standardization or robust scaling would need to account for.

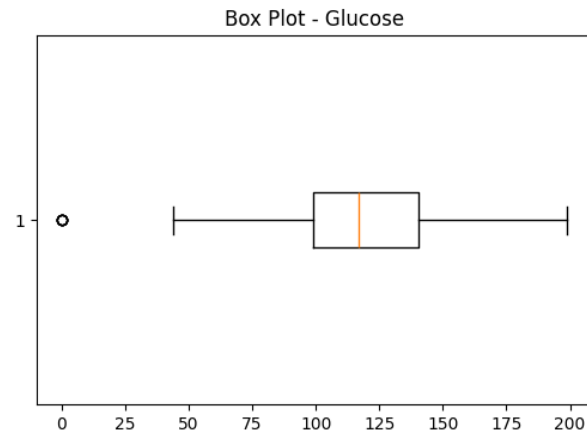


Figure 48: Box Plot of Glucose – Pima Indians Diabetes Dataset

Inference: The IQR-based outlier check flagged 5 value(s) in Glucose lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Glucose that standardization or robust scaling would need to account for.

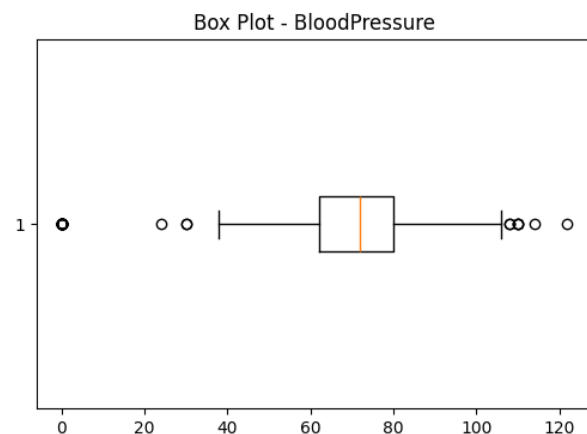


Figure 49: Box Plot of BloodPressure – Pima Indians Diabetes Dataset

Inference: The IQR-based outlier check flagged 45 value(s) in BloodPressure lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of BloodPressure that standardization or robust scaling would need to account for.

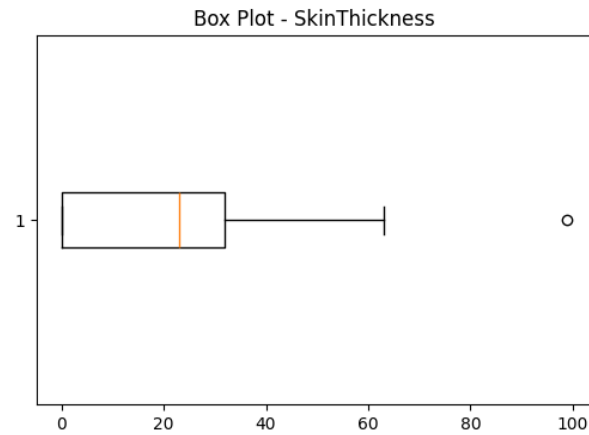


Figure 50: Box Plot of SkinThickness – Pima Indians Diabetes Dataset

Inference: The IQR-based outlier check flagged 1 value(s) in SkinThickness lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of SkinThickness that standardization or robust scaling would need to account for.

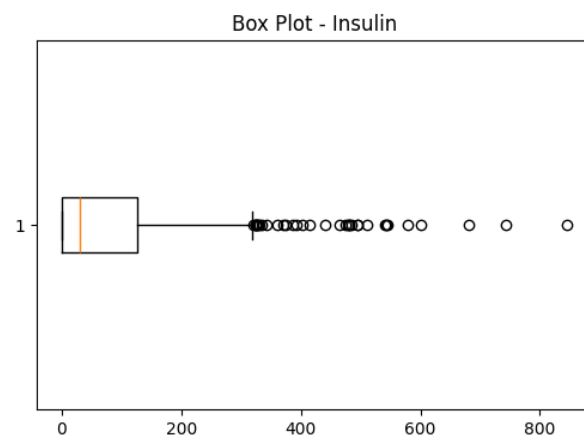


Figure 51: Box Plot of Insulin – Pima Indians Diabetes Dataset

Inference: The IQR-based outlier check flagged 34 value(s) in Insulin lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Insulin that standardization or robust scaling would need to account for.

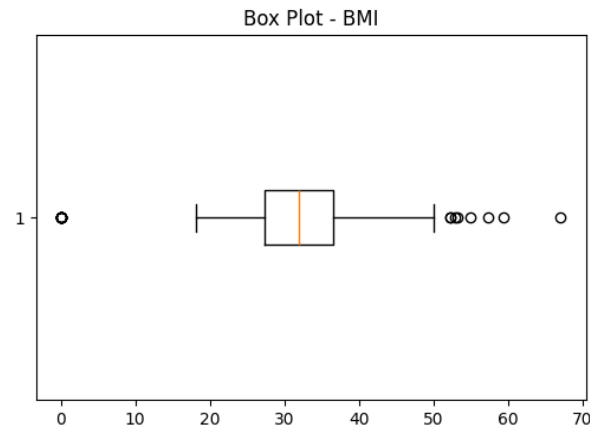


Figure 52: Box Plot of BMI – Pima Indians Diabetes Dataset

Inference: The IQR-based outlier check flagged 19 value(s) in BMI lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of BMI that standardization or robust scaling would need to account for.

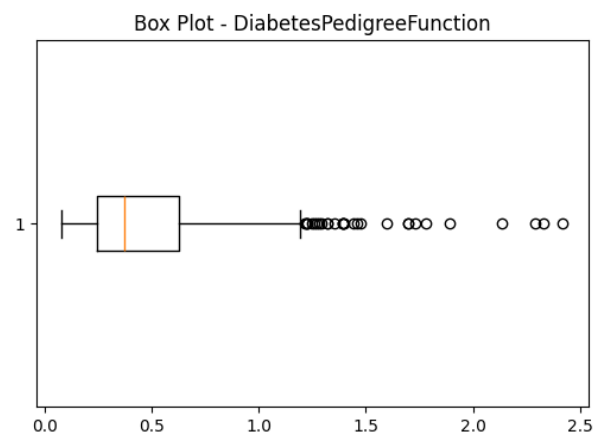


Figure 53: Box Plot of DiabetesPedigreeFunction – Pima Indians Diabetes Dataset

Inference: The IQR-based outlier check flagged 29 value(s) in DiabetesPedigreeFunction lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of DiabetesPedigreeFunction that standardization or robust scaling would need to account for.

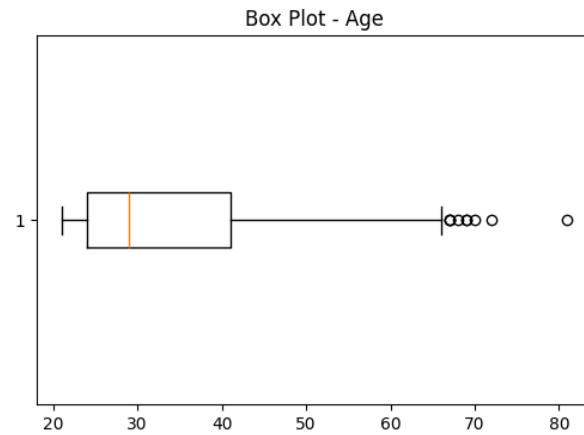


Figure 54: Box Plot of Age – Pima Indians Diabetes Dataset

Inference: The IQR-based outlier check flagged 9 value(s) in Age lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Age that standardization or robust scaling would need to account for.

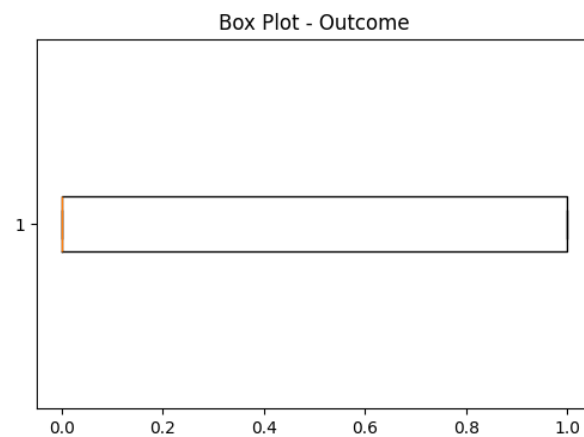


Figure 55: Box Plot of Outcome – Pima Indians Diabetes Dataset

Inference: No values in Outcome were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

Bar Charts – Pima Indians Diabetes Dataset

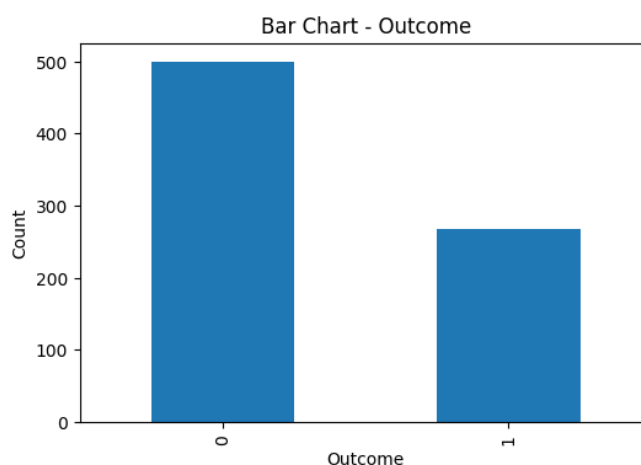


Figure 56: Bar Chart of Outcome – Pima Indians Diabetes Dataset

Inference: non-diabetic cases (Outcome = 0) considerably outnumber diabetic cases (Outcome = 1), with roughly 65% negative and 35% positive, indicating a moderately imbalanced binary target.

Scatter Plot – Pima Indians Diabetes Dataset

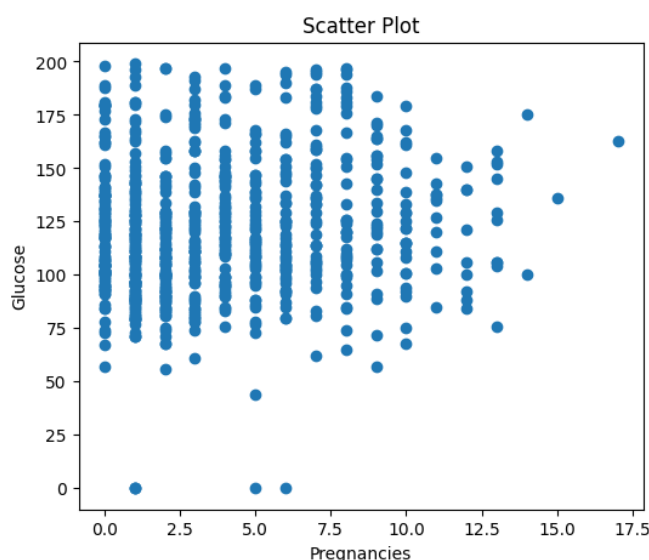


Figure 57: Scatter Plot of Pregnancies vs Glucose – Pima Indians Diabetes Dataset

Inference: Pregnancies and Glucose show a loosely scattered relationship with no strong linear trend, though the density of points is greatest at lower pregnancy counts across the full glucose range, consistent with Glucose being a stronger individual predictor of diabetes than Pregnancies.

Correlation Heatmap – Pima Indians Diabetes Dataset

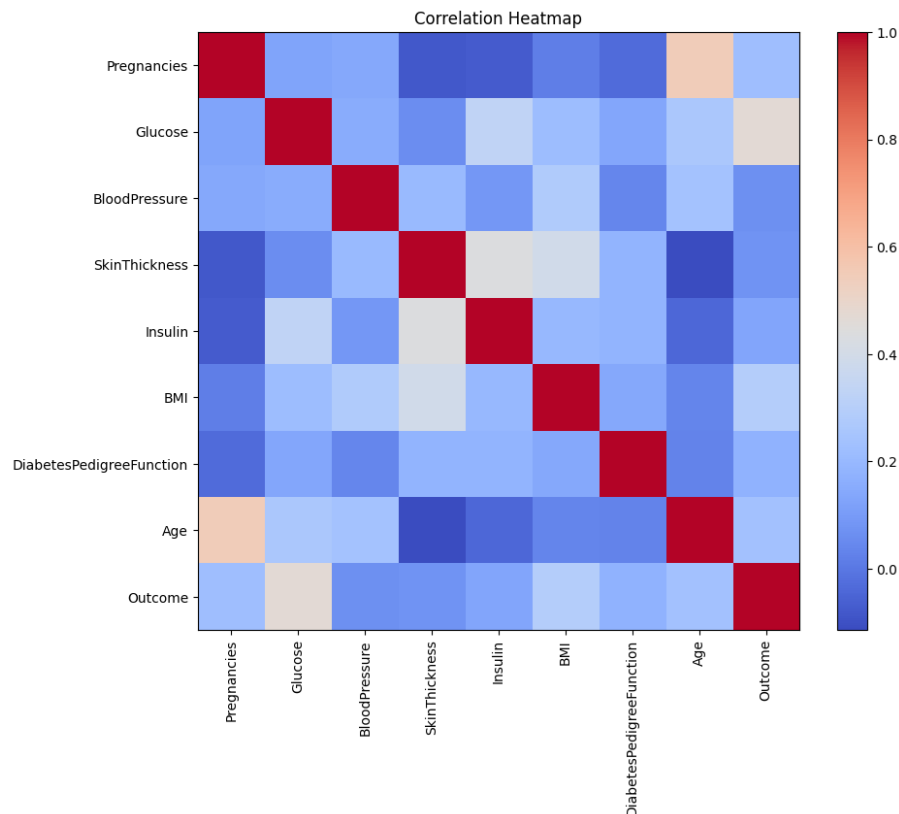


Figure 58: Correlation Heatmap – Pima Indians Diabetes Dataset

Inference: Glucose shows the strongest correlation with Outcome among all features, consistent with its established clinical role in diabetes diagnosis, while BMI and Age show weaker but still positive associations with the target. Most feature pairs otherwise show low to moderate correlation, suggesting limited redundancy among predictors.

Iris Dataset

The raw Iris Dataset contains 150 rows and 6 columns, with 0 duplicate row(s) detected via `df.duplicated().sum()`. No missing values were detected in this dataset, so no imputation was required.

Histograms – Iris Dataset

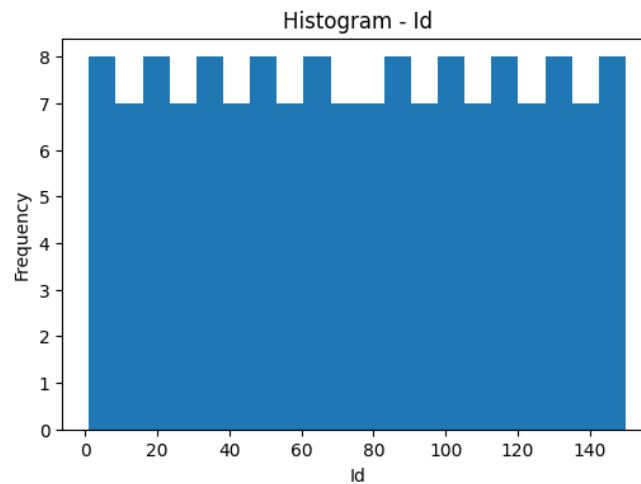


Figure 59: Histogram of Id – Iris Dataset

Inference: Id is a simple row index from 1 to 150 and therefore uniformly distributed with no analytical value. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

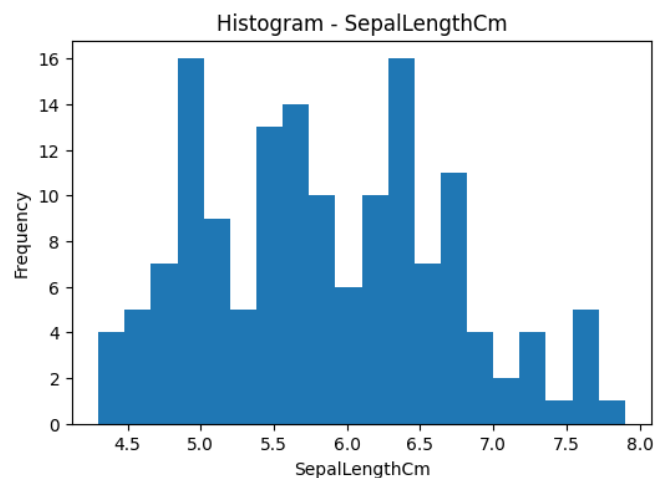


Figure 60: Histogram of SepalLengthCm – Iris Dataset

Inference: SepalLengthCm ranges from 4.3 to 7.9 cm with a mean of 5.84 and a standard deviation of 0.83, showing a fairly symmetric, mildly multi-modal distribution across the three species. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

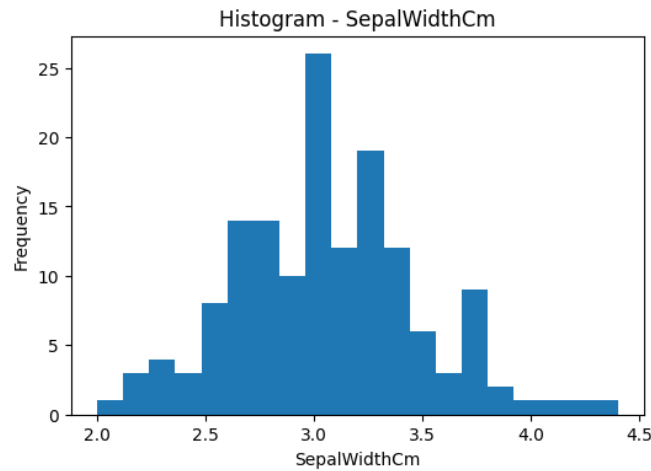


Figure 61: Histogram of SepalWidthCm – Iris Dataset

Inference: SepalWidthCm ranges from 2.0 to 4.4 cm with a mean of 3.05 and a standard deviation of 0.43, and is the only feature in this dataset with detected outliers (4 by the IQR rule). Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

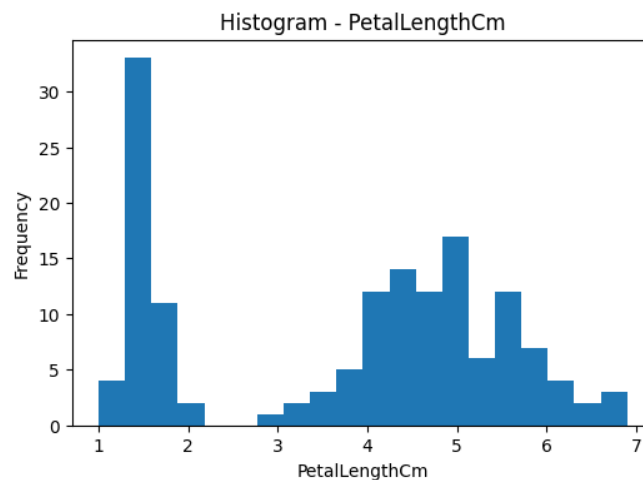


Figure 62: Histogram of PetalLengthCm – Iris Dataset

Inference: PetalLengthCm ranges from 1.0 to 6.9 cm with a mean of 3.76 and a standard deviation of 1.76, showing a clearly bimodal shape driven by the separation between Iris-setosa and the other two species. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

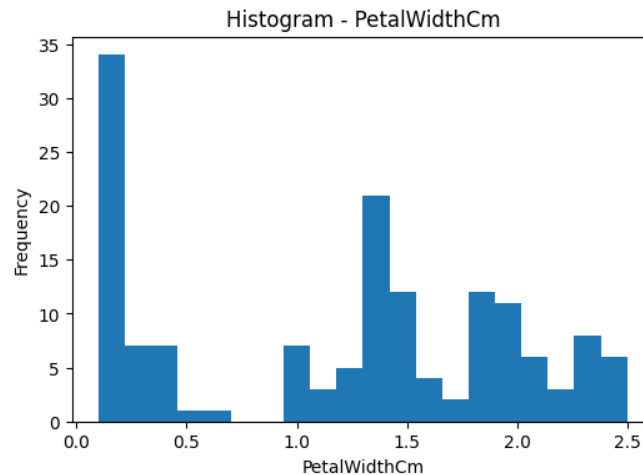


Figure 63: Histogram of PetalWidthCm – Iris Dataset

Inference: PetalWidthCm ranges from 0.1 to 2.5 cm with a mean of 1.20 and a standard deviation of 0.76, mirroring the bimodal pattern seen in petal length. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

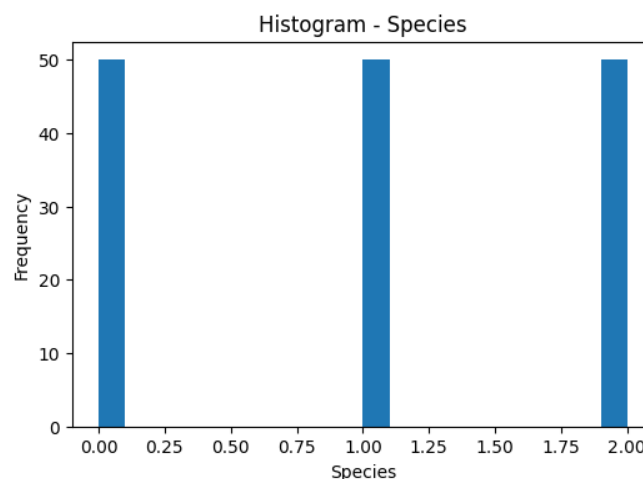


Figure 64: Histogram of Species – Iris Dataset

Inference: Species the encoded target label with three balanced classes, each containing exactly 50 samples. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

Box Plots – Iris Dataset

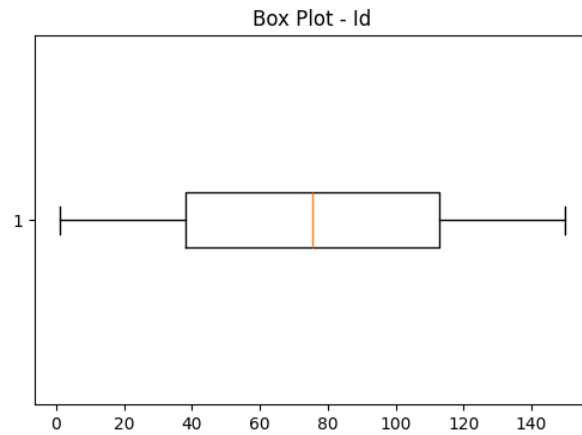


Figure 65: Box Plot of Id – Iris Dataset

Inference: No values in Id were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

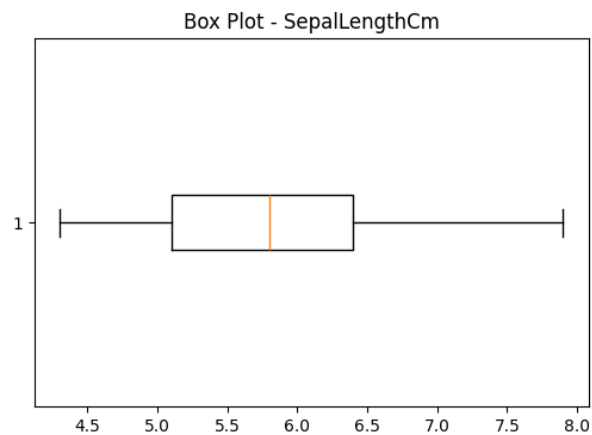


Figure 66: Box Plot of SepalLengthCm – Iris Dataset

Inference: No values in SepalLengthCm were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

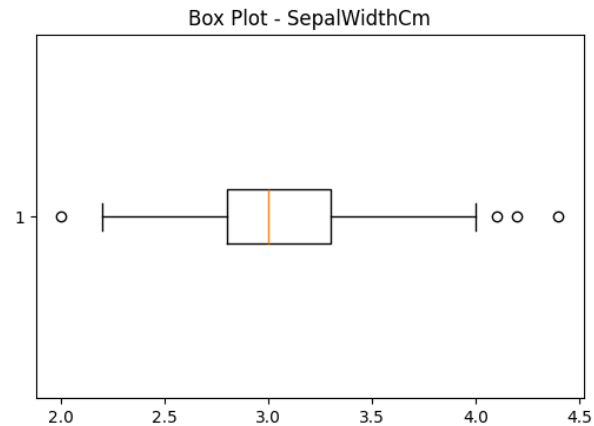


Figure 67: Box Plot of SepalWidthCm – Iris Dataset

Inference: The IQR-based outlier check flagged 4 value(s) in SepalWidthCm lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of SepalWidthCm that standardization or robust scaling would need to account for.

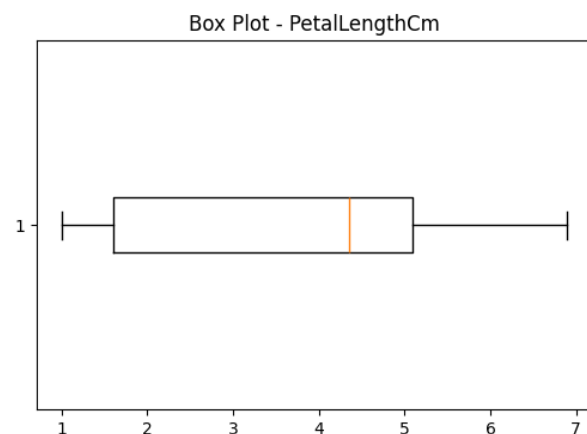


Figure 68: Box Plot of PetalLengthCm – Iris Dataset

Inference: No values in PetalLengthCm were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

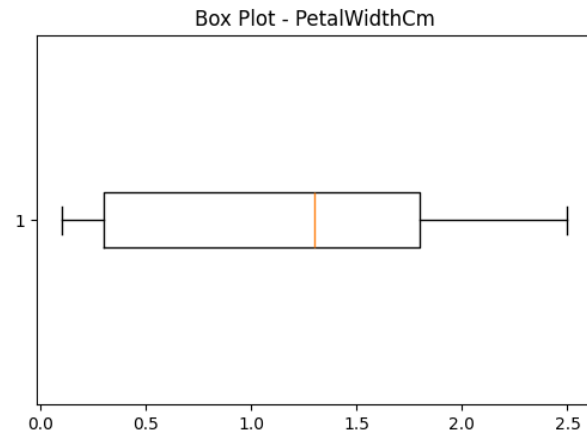


Figure 69: Box Plot of PetalWidthCm – Iris Dataset

Inference: No values in PetalWidthCm were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

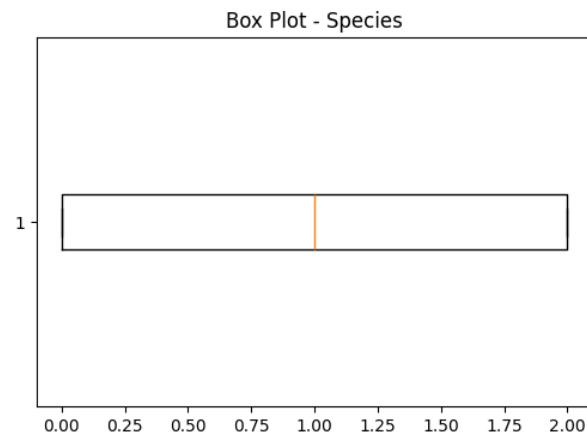


Figure 70: Box Plot of Species – Iris Dataset

Inference: No values in Species were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

Bar Charts – Iris Dataset

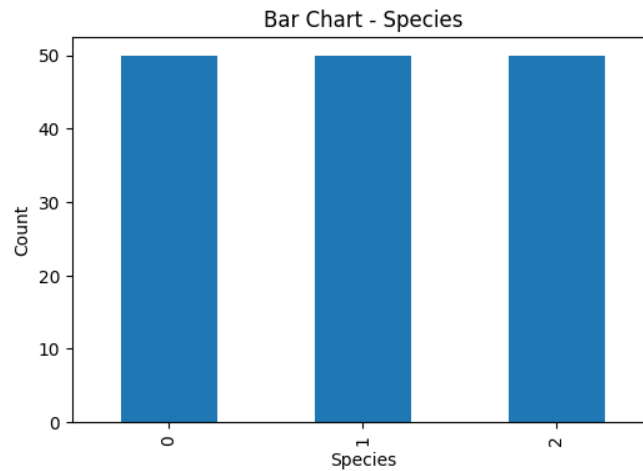


Figure 71: Bar Chart of Species – Iris Dataset

Inference: the three species – Iris-setosa, Iris-versicolor and Iris-virginica – are perfectly balanced with 50 samples each, making this an ideal dataset for multi-class classification without any class-imbalance concerns.

Scatter Plot – Iris Dataset

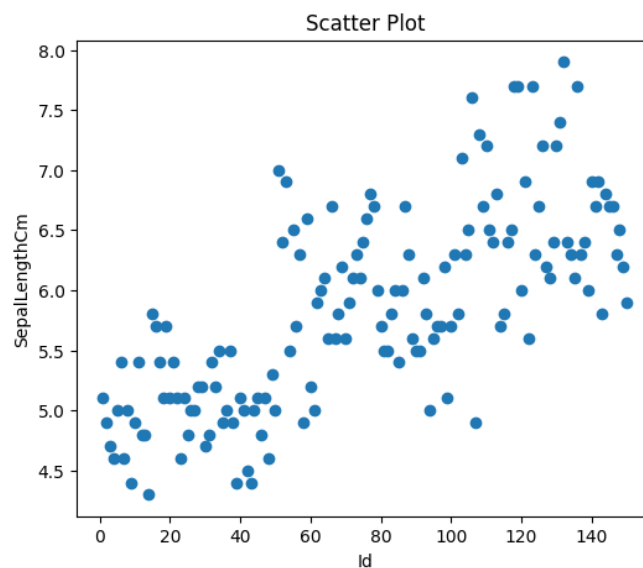


Figure 72: Scatter Plot of Id vs SepalLengthCm – Iris Dataset

Inference: Because the row identifier Id is plotted on the x-axis, the points fall into three horizontal bands corresponding to the three species blocks in the file, within which SepalLengthCm varies around a fairly consistent range – PetalLengthCm vs PetalWidthCm would show the classic, much more visually striking Iris species separation.

Correlation Heatmap – Iris Dataset

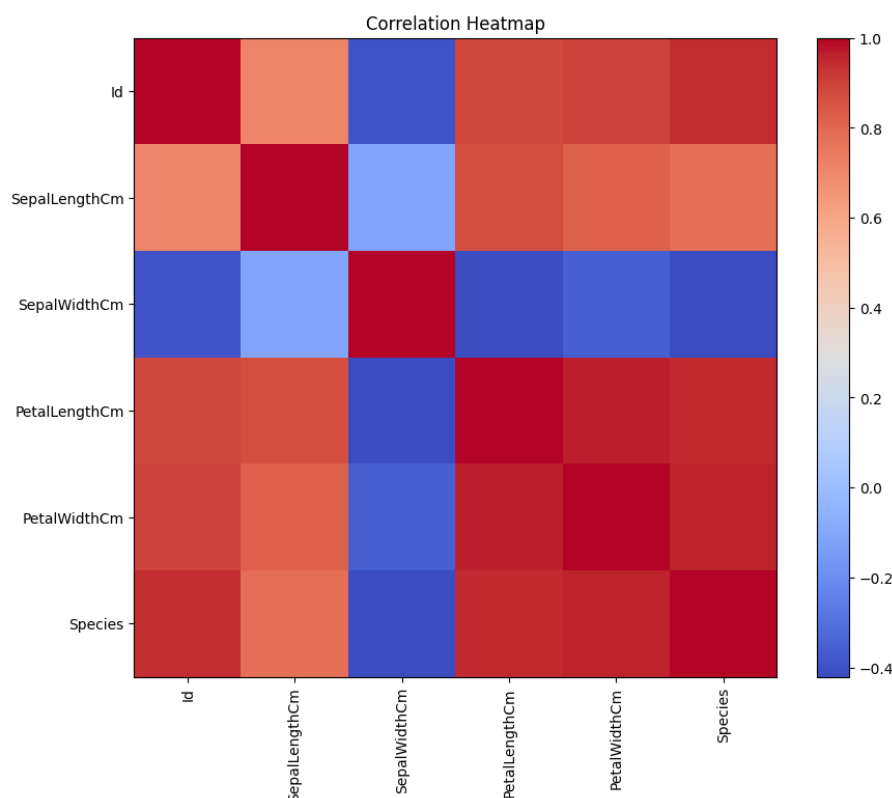


Figure 73: Correlation Heatmap – Iris Dataset

Inference: PetalLengthCm and PetalWidthCm show a very strong positive correlation with each other and with SepalLengthCm, reflecting the well-known fact that petal measurements are highly informative for distinguishing Iris species, while SepalWidthCm behaves almost independently of the other three measurements.

SMS Spam Collection Dataset

The raw SMS Spam Collection Dataset contains 5572 rows and 5 columns, with 403 duplicate row(s) detected via `df.duplicated().sum()`.

Table 4: Missing Values – SMS Spam Collection Dataset

Column	Missing Values	Percentage (%)
Unnamed: 2	5522	99.10
Unnamed: 3	5560	99.78
Unnamed: 4	5566	99.89

Missing values were imputed using the column mode for categorical fields and the column median for numerical fields, after which no missing values remained in the dataset.

Histograms – SMS Spam Collection Dataset

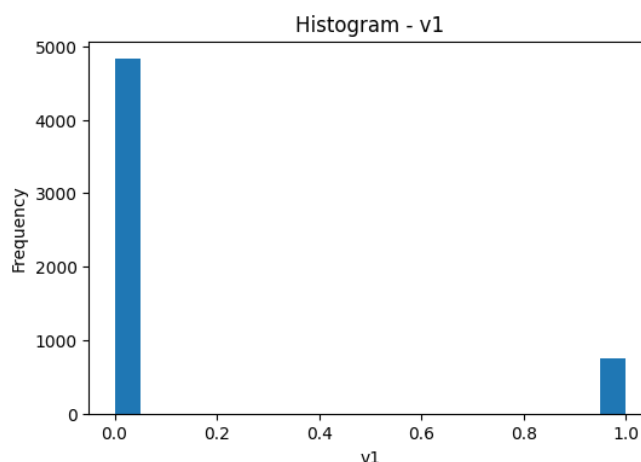


Figure 74: Histogram of v1 – SMS Spam Collection Dataset

Inference: v1 the encoded ham/spam label; 4825 of 5572 messages (86.6%) are ham, so the raw dataset is considerably imbalanced toward legitimate messages. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

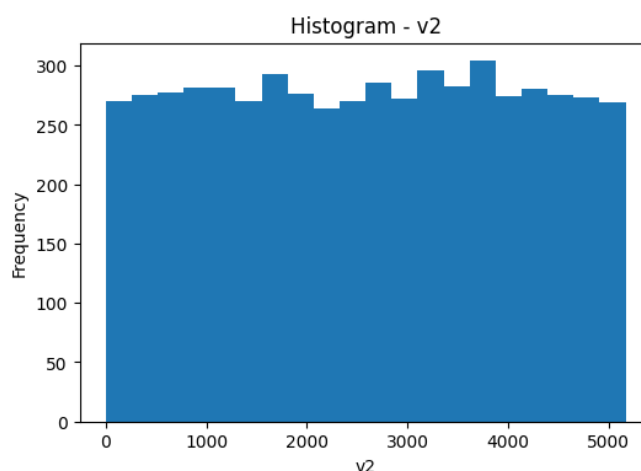


Figure 75: Histogram of v2 – SMS Spam Collection Dataset

Inference: v2 the encoded message text column; with 5169 unique messages out of 5572 rows, almost every value is distinct, so its histogram is effectively a near-uniform spread across encoded index values. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

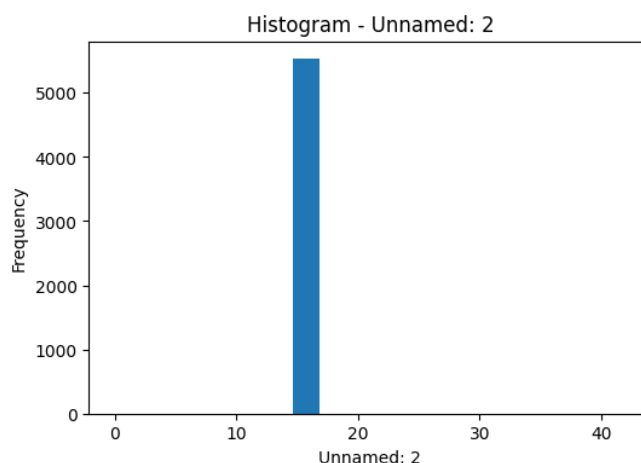


Figure 76: Histogram of Unnamed: 2 – SMS Spam Collection Dataset

Inference: Unnamed: 2 an almost entirely empty artefact column (99.1% missing before imputation) that captures stray comma-separated text from a small number of malformed rows. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

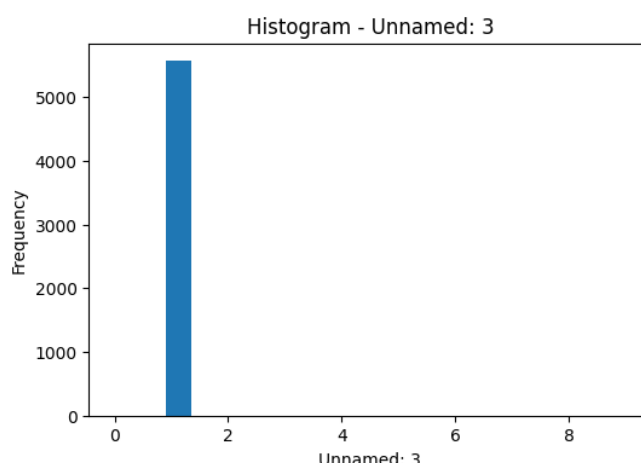


Figure 77: Histogram of Unnamed: 3 – SMS Spam Collection Dataset

Inference: Unnamed: 3 an even sparser artefact column (99.8% missing before imputation), also filled with mode-imputed placeholder text. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

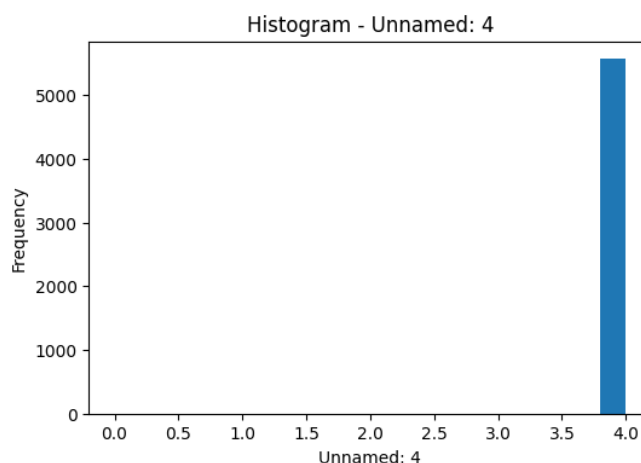


Figure 78: Histogram of Unnamed: 4 – SMS Spam Collection Dataset

Inference: Unnamed: 4 the sparsest of the three artefact columns (99.9% missing before imputation). Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

Box Plots – SMS Spam Collection Dataset

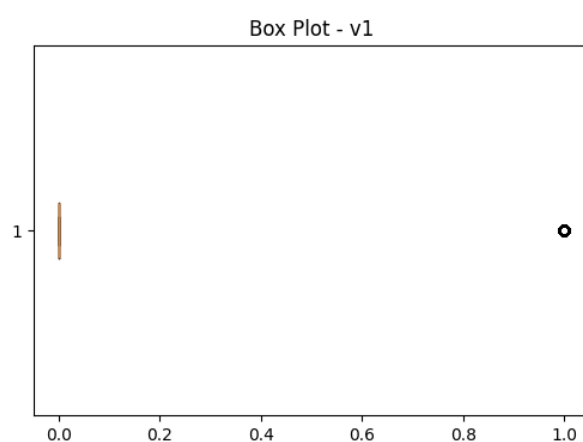


Figure 79: Box Plot of v1 – SMS Spam Collection Dataset

Inference: The IQR-based outlier check flagged 747 value(s) in v1 lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of v1 that standardization or robust scaling would need to account for.

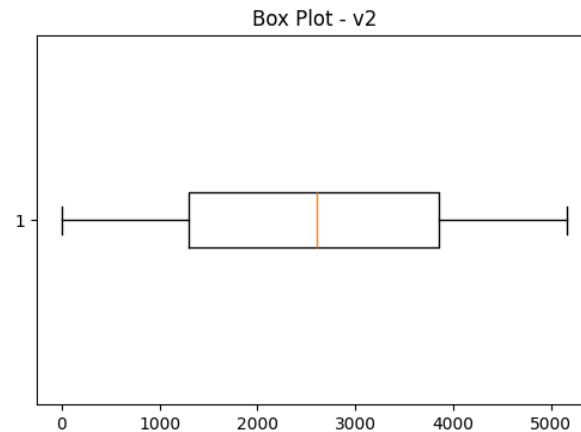


Figure 80: Box Plot of v2 – SMS Spam Collection Dataset

Inference: No values in v2 were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

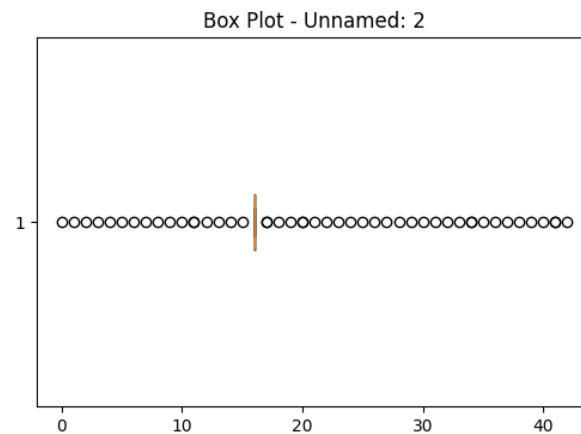


Figure 81: Box Plot of Unnamed: 2 – SMS Spam Collection Dataset

Inference: The IQR-based outlier check flagged 47 value(s) in Unnamed: 2 lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Unnamed: 2 that standardization or robust scaling would need to account for.

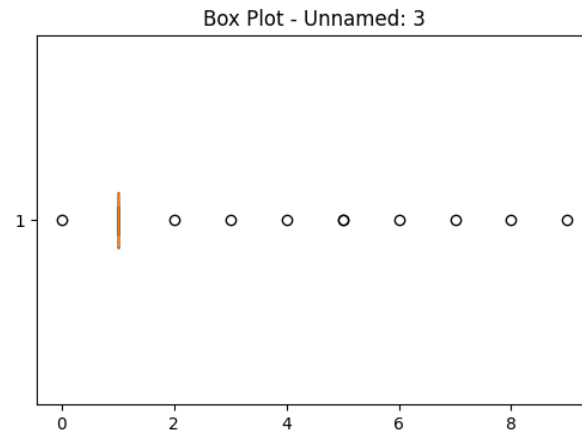


Figure 82: Box Plot of Unnamed: 3 – SMS Spam Collection Dataset

Inference: The IQR-based outlier check flagged 10 value(s) in Unnamed: 3 lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Unnamed: 3 that standardization or robust scaling would need to account for.

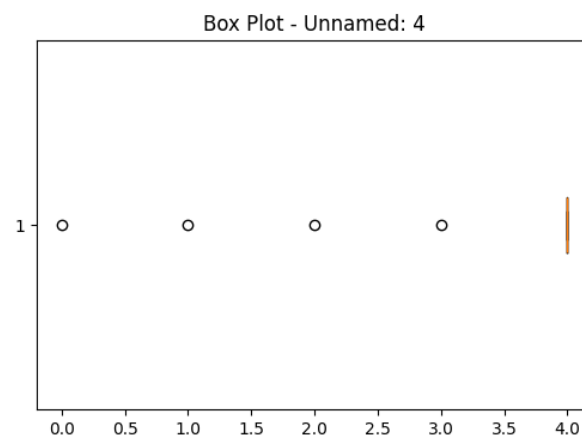


Figure 83: Box Plot of Unnamed: 4 – SMS Spam Collection Dataset

Inference: The IQR-based outlier check flagged 4 value(s) in Unnamed: 4 lying beyond 1.5 times the interquartile range. Together with the distribution noted above, this indicates the spread and tail behaviour of Unnamed: 4 that standardization or robust scaling would need to account for.

Bar Charts – SMS Spam Collection Dataset

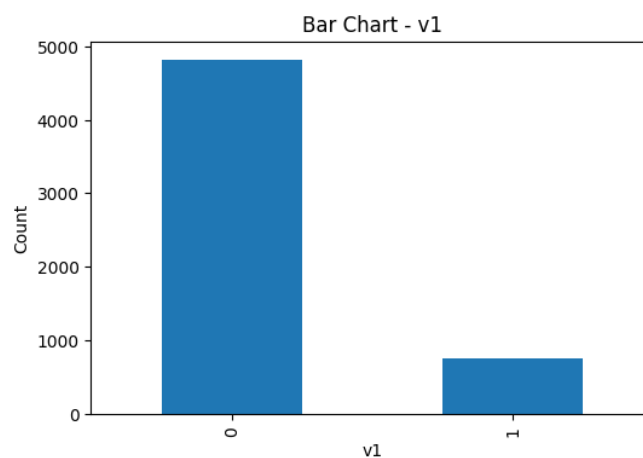


Figure 84: Bar Chart of v1 – SMS Spam Collection Dataset

Inference: ham messages (4825) dominate the dataset compared to spam messages (747), giving roughly an 86:14 class split that would need to be accounted for when training a spam classifier.

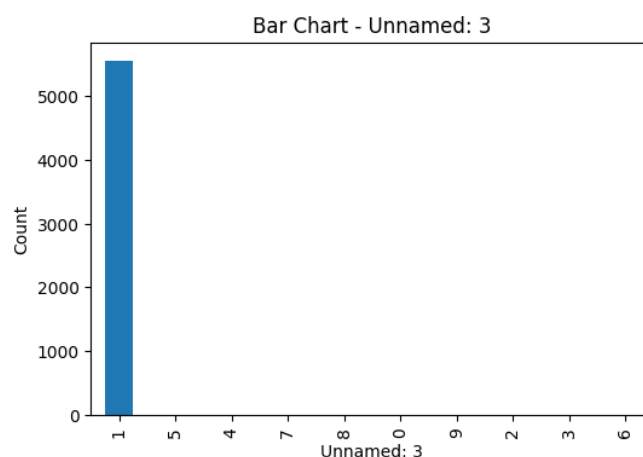


Figure 85: Bar Chart of Unnamed: 3 – SMS Spam Collection Dataset

Inference: after mode-imputation this artefact column collapses almost entirely into a single dominant category, since fewer than 1% of rows originally held any real value here.

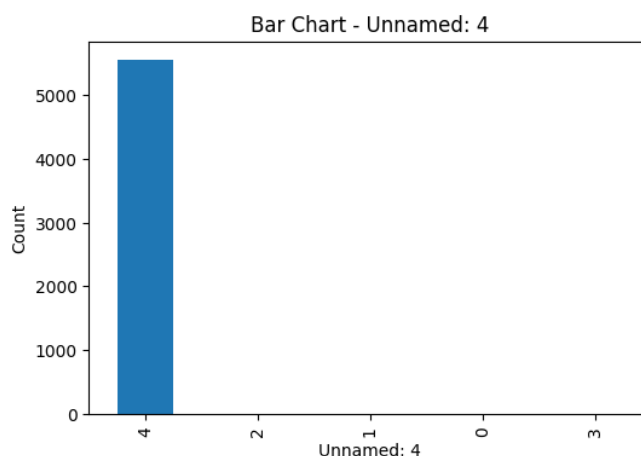


Figure 86: Bar Chart of Unnamed: 4 – SMS Spam Collection Dataset

Inference: similarly, this column is dominated by a single imputed category, reflecting how sparse the original data was (only 6 non-null values out of 5572 rows).

Scatter Plot – SMS Spam Collection Dataset

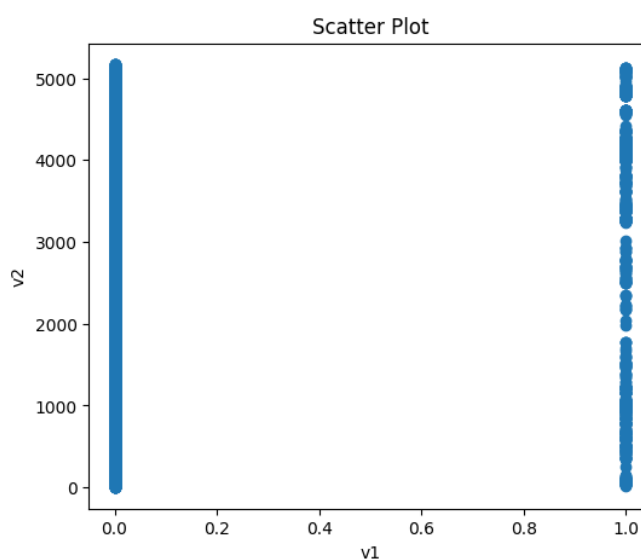


Figure 87: Scatter Plot of v1 vs v2 – SMS Spam Collection Dataset

Inference: With v1 restricted to two encoded values (ham/spam) and v2 spread across thousands of encoded text indices, the scatter plot forms two dense vertical strips rather than a meaningful trend, underscoring that this dataset's real signal lies in the text content itself rather than in these two encoded columns viewed jointly.

Correlation Heatmap – SMS Spam Collection Dataset

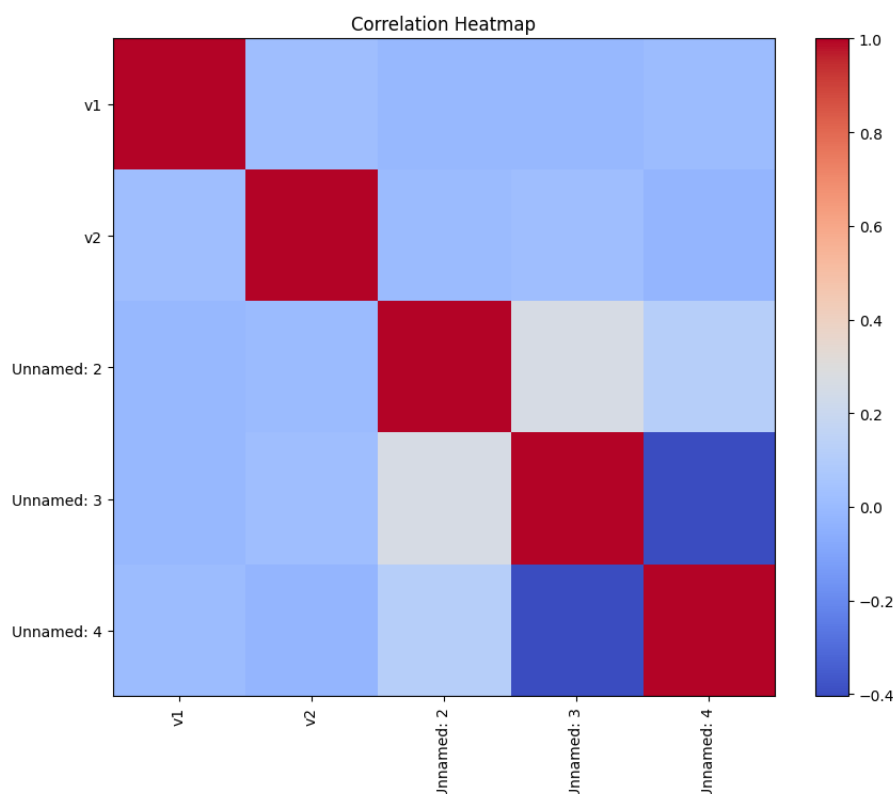


Figure 88: Correlation Heatmap – SMS Spam Collection Dataset

Inference: Correlations among the five encoded columns are weak overall, which is expected since v2 is an encoding of largely unique free-text messages and the three unnamed columns are mostly imputed placeholder values; v1 (the spam/ham label) shows only a marginal relationship with the other encoded columns, reinforcing that meaningful spam-detection features would need to come from text-derived attributes such as word frequencies rather than these raw columns.

MNIST Handwritten Digit Dataset (train split)

The raw MNIST Handwritten Digit Dataset (train split) contains 60000 rows and 785 columns, with 0 duplicate row(s) detected via `df.duplicated().sum()`. No missing values were detected in this dataset, so no imputation was required.

Histograms – MNIST Handwritten Digit Dataset (train split)

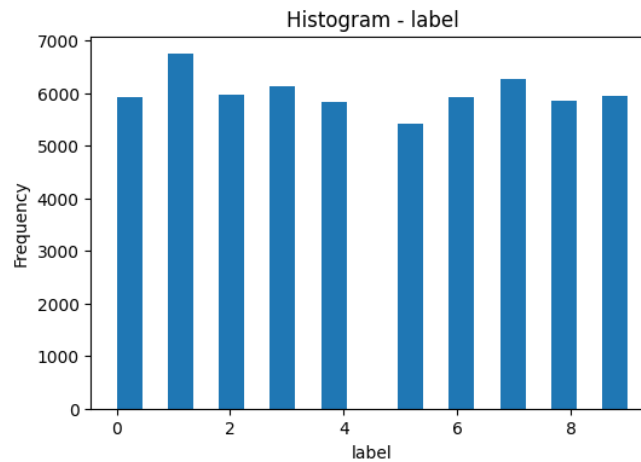


Figure 89: Histogram of label – MNIST Handwritten Digit Dataset (train split)

Inference: label the digit class (0-9) with a mean of 4.45 and a standard deviation of 2.89 across 60000 training images, reasonably balanced across all ten digits. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

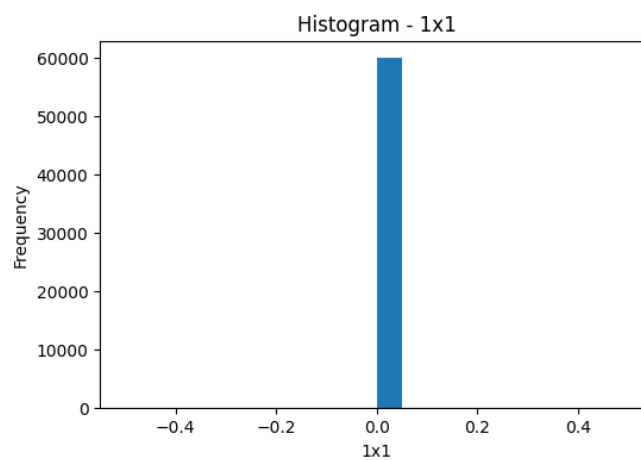


Figure 90: Histogram of 1x1 – MNIST Handwritten Digit Dataset (train split)

Inference: 1x1 corresponds to the top-left corner pixel of each 28x28 image; every one of the 60000 samples has a value of exactly 0, since handwritten digits never extend into this corner. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

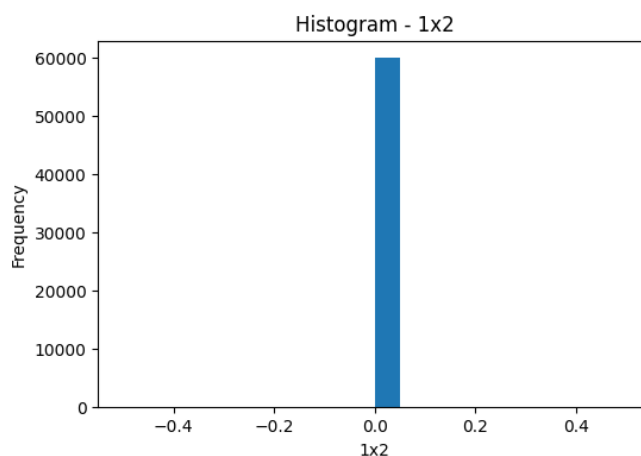


Figure 91: Histogram of 1x2 – MNIST Handwritten Digit Dataset (train split)

Inference: 1x2 another top-row border pixel that is constant at 0 across the entire training set. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

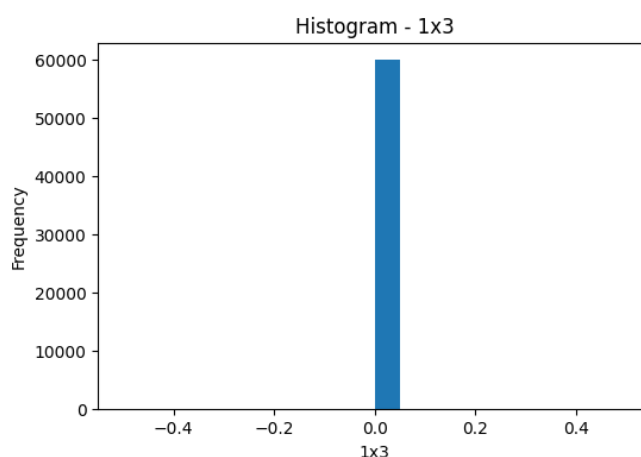


Figure 92: Histogram of 1x3 – MNIST Handwritten Digit Dataset (train split)

Inference: 1x3 a top-row border pixel that remains 0 for all 60,000 samples. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

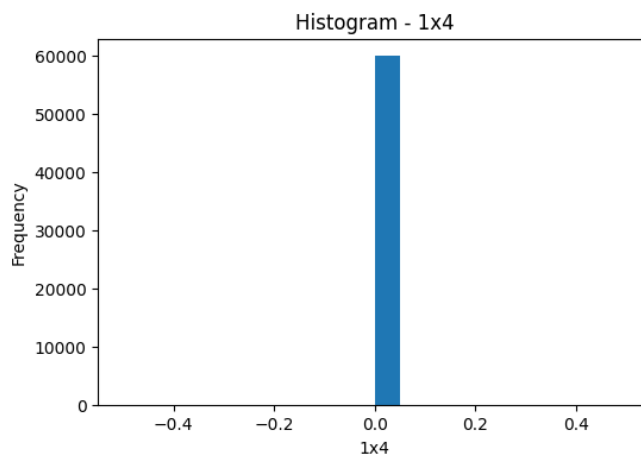


Figure 93: Histogram of 1x4 – MNIST Handwritten Digit Dataset (train split)

Inference: 1x4 a top-row border pixel that is uniformly 0 across the dataset. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

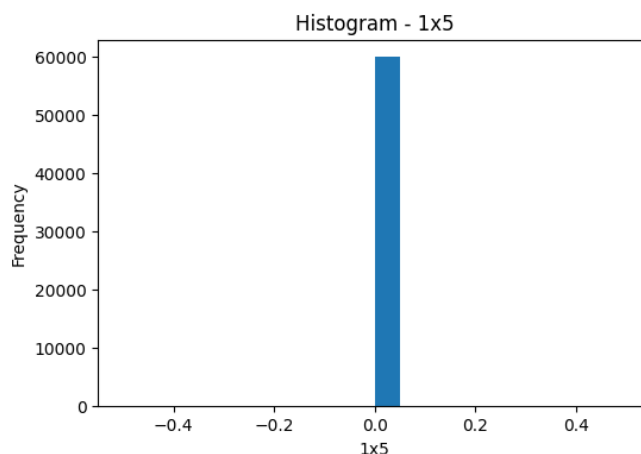


Figure 94: Histogram of 1x5 – MNIST Handwritten Digit Dataset (train split)

Inference: 1x5 a top-row border pixel that shows no variation (constant 0) in this split. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

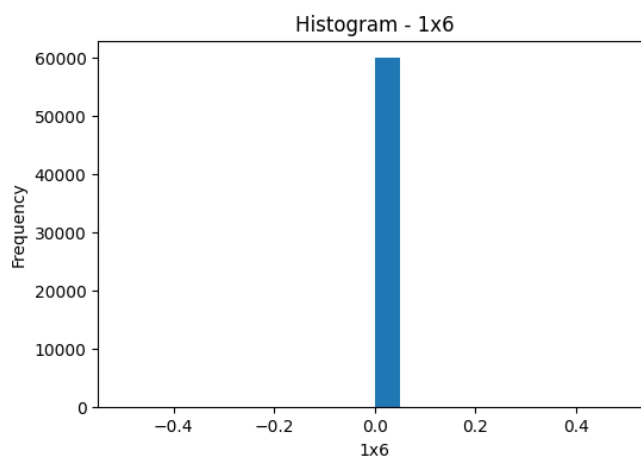


Figure 95: Histogram of 1x6 – MNIST Handwritten Digit Dataset (train split)

Inference: 1x6 a top-row border pixel that stays at 0 for every training sample. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

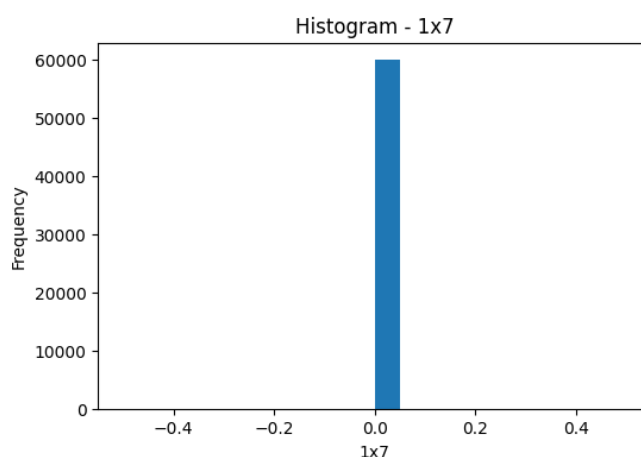


Figure 96: Histogram of 1x7 – MNIST Handwritten Digit Dataset (train split)

Inference: 1x7 a top-row border pixel with zero variance across all 60,000 rows. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

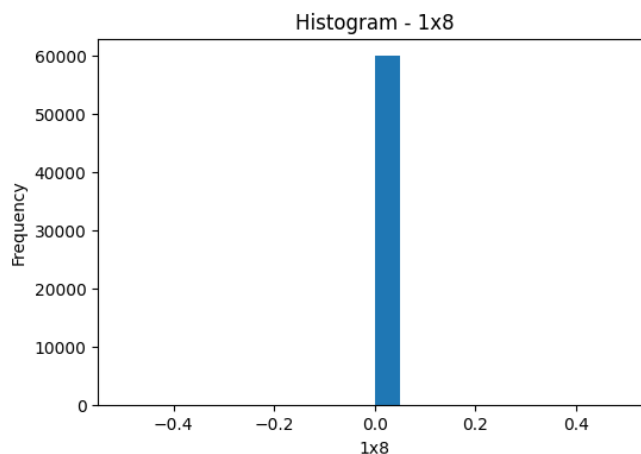


Figure 97: Histogram of 1x8 – MNIST Handwritten Digit Dataset (train split)

Inference: 1x8 a top-row border pixel that remains constant at 0 throughout the dataset. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

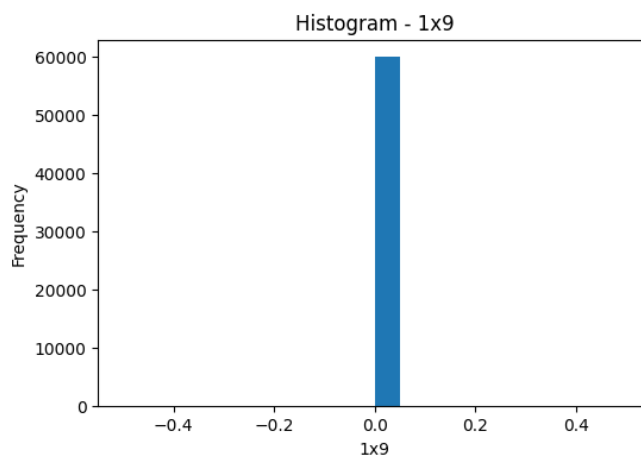


Figure 98: Histogram of 1x9 – MNIST Handwritten Digit Dataset (train split)

Inference: 1x9 a top-row border pixel that is uniformly 0, like the eight pixels preceding it. Examining its histogram alongside the other features helps establish whether the variable needs transformation or scaling before it is used in a downstream model.

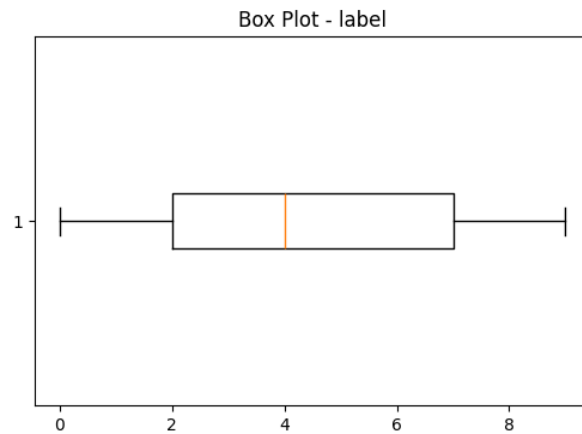
Box Plots – MNIST Handwritten Digit Dataset (train split)

Figure 99: Box Plot of label – MNIST Handwritten Digit Dataset (train split)

Inference: No values in label were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

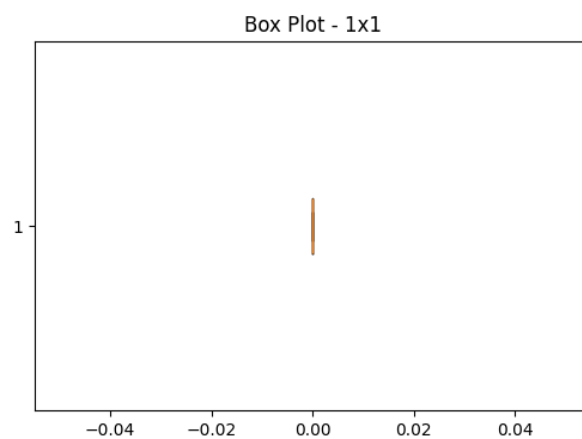


Figure 100: Box Plot of 1x1 – MNIST Handwritten Digit Dataset (train split)

Inference: No values in 1x1 were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

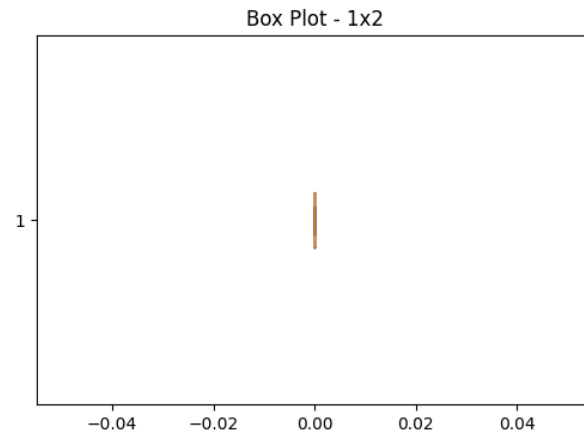


Figure 101: Box Plot of 1x2 – MNIST Handwritten Digit Dataset (train split)

Inference: No values in 1x2 were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

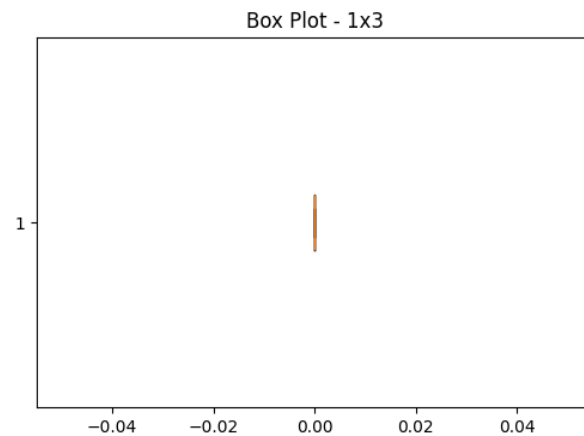


Figure 102: Box Plot of 1x3 – MNIST Handwritten Digit Dataset (train split)

Inference: No values in 1x3 were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

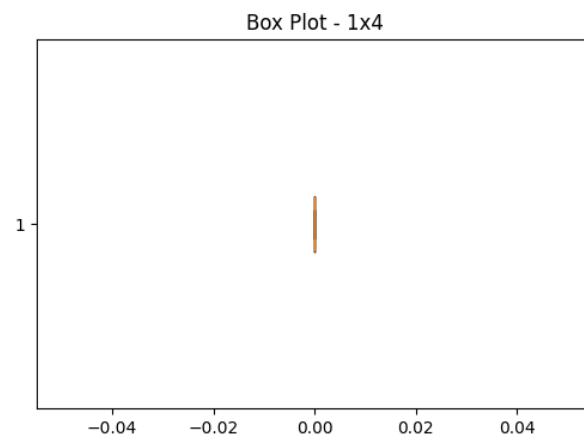


Figure 103: Box Plot of 1x4 – MNIST Handwritten Digit Dataset (train split)

Inference: No values in 1x4 were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

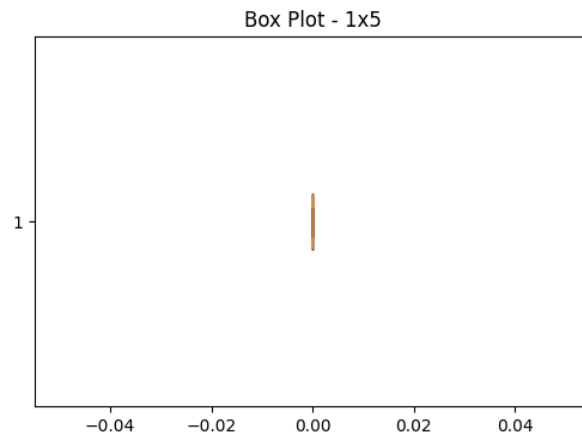


Figure 104: Box Plot of 1x5 – MNIST Handwritten Digit Dataset (train split)

Inference: No values in 1x5 were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

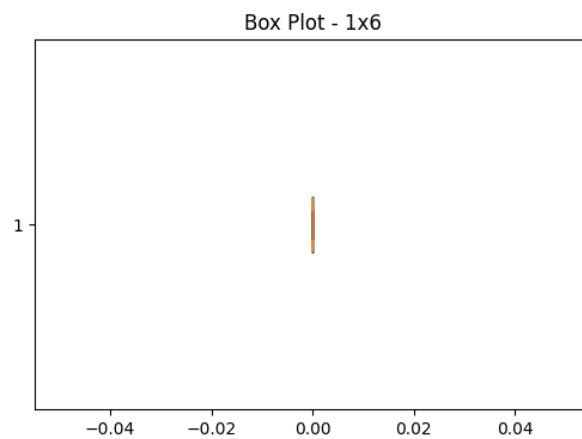


Figure 105: Box Plot of 1x6 – MNIST Handwritten Digit Dataset (train split)

Inference: No values in 1x6 were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

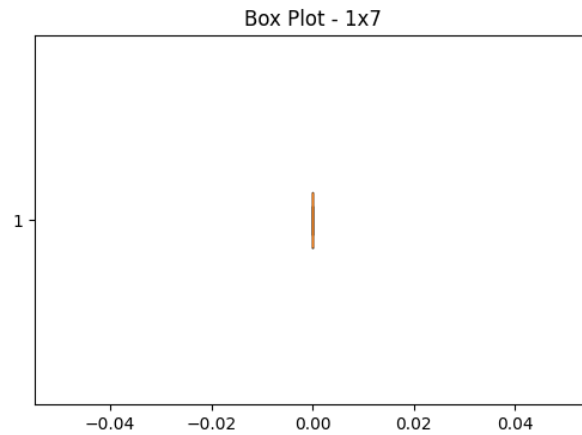


Figure 106: Box Plot of 1x7 – MNIST Handwritten Digit Dataset (train split)

Inference: No values in 1x7 were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

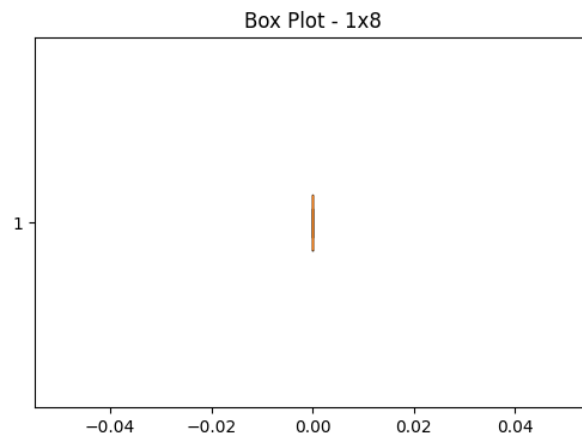


Figure 107: Box Plot of 1x8 – MNIST Handwritten Digit Dataset (train split)

Inference: No values in 1x8 were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

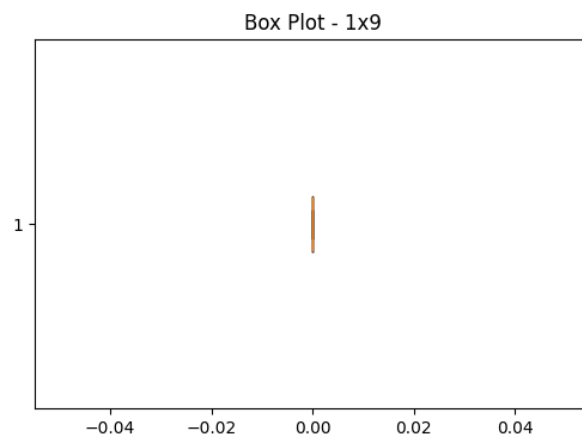


Figure 108: Box Plot of 1x9 – MNIST Handwritten Digit Dataset (train split)

Inference: No values in 1x9 were flagged as outliers by the $1.5 \times \text{IQR}$ rule, indicating the column is relatively free of extreme values within this dataset.

Bar Charts – MNIST Handwritten Digit Dataset (train split)

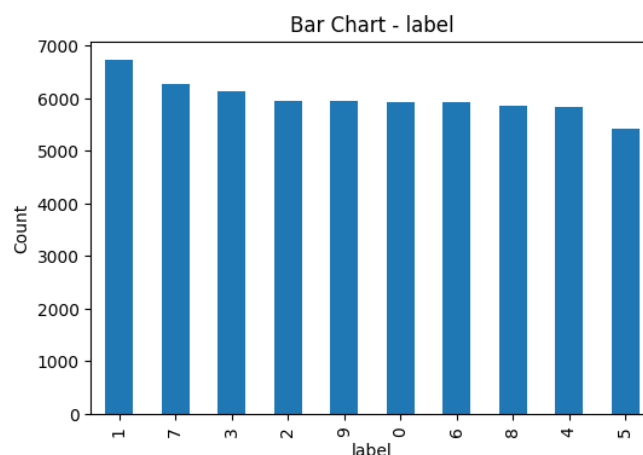


Figure 109: Bar Chart of label – MNIST Handwritten Digit Dataset (train split)

Inference: the ten digit classes (0-9) are reasonably well balanced, each accounting for roughly 9-11% of the 60000 training images, making MNIST well suited to standard multi-class classification without heavy resampling.

Scatter Plot – MNIST Handwritten Digit Dataset (train split)

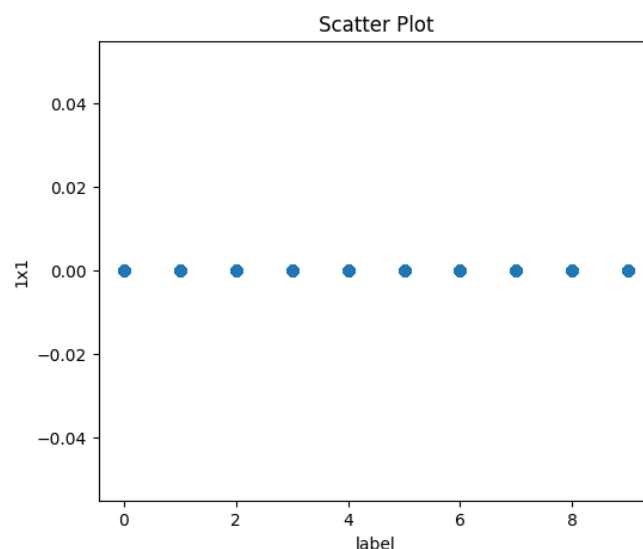


Figure 110: Scatter Plot of label vs 1x1 – MNIST Handwritten Digit Dataset (train split)

Inference: Because pixel 1x1 is constant at 0 for every single image in the training set, the scatter plot collapses to a flat horizontal line spanning the ten digit labels – a reminder that the outer border pixels of MNIST images carry no discriminative information, and that the informative pixels lie toward the centre of the 28x28 grid.

Correlation Heatmap – MNIST Handwritten Digit Dataset (train split)

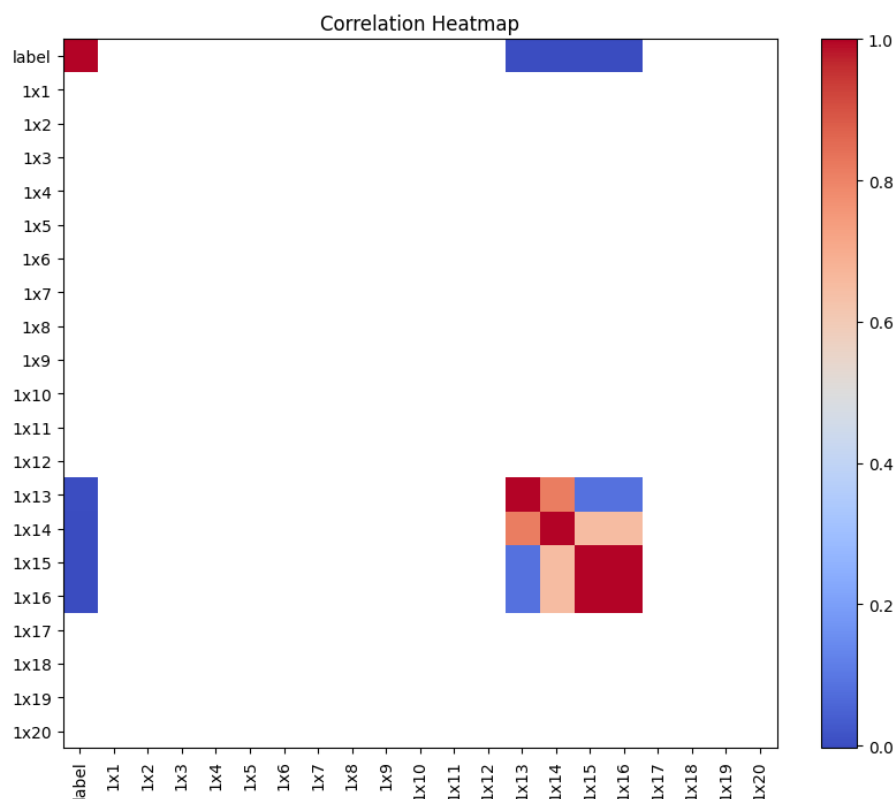


Figure 111: Correlation Heatmap – MNIST Handwritten Digit Dataset (train split)

Inference: The heatmap covers the label column together with the first 20 pixel columns (1x1 through 20x1 in row-major order), most of which sit along the top border of the image and are constant or near-constant at 0; consequently the heatmap is dominated by near-zero correlations, and only a handful of interior-adjacent pixels show any mild positive correlation with each other.

8 Data Preprocessing

The same preprocessing pattern was applied to all five datasets, implemented as a short loop over every column of the DataFrame:

```

1 for col in df.columns:
2     if df[col].dtype == "object":
3         df[col] = df[col].fillna(df[col].mode()[0])
4     else:
5         df[col] = df[col].fillna(df[col].median())
6
7 label_encoder = LabelEncoder()
8 for col in df.select_dtypes(include="object").columns:
9     df[col] = label_encoder.fit_transform(df[col])

```

- **Missing Value Handling:** Object-typed (categorical) columns were imputed with the column mode, and numeric columns with the column median. This affected seven columns in the Loan dataset (ranging from 0.49% missing in `Married` to 8.14% in `Credit_History`) and three columns in the Spam dataset (the `Unnamed: 2/3/4` artefact columns, each over 99% missing). The Diabetes, Iris and MNIST datasets had no missing values at all.
- **Dropping Irrelevant Features:** The notebook does not explicitly drop any columns before encoding; instead, columns of little predictive value (such as `Loan_ID` in the Loan dataset and `Id` in the Iris dataset) are retained but label-encoded along with everything else, since they are object/identifier columns. In a downstream modelling stage these identifier columns would typically be dropped rather than encoded.
- **Encoding Categorical Variables:** Every object-typed column was label-encoded using `LabelEncoder`, converting text categories (e.g., `Male/Female`, `ham/spam`, `Iris-setosa/Iris-versicolor/Iris-virginica`) into integer codes so that every column in each `DataFrame` becomes numeric.
- **Outlier Detection:** For every numerical column, the interquartile range (IQR) rule (values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$) was used to count outliers. The counts are reported alongside each box plot in Section 7 and summarized dataset-by-dataset in Section 11.
- **Standardization and Normalization:** After encoding, every dataset's numerical columns were standardized with `StandardScaler` (zero mean, unit variance) and separately normalized with `MinMaxScaler` (scaled to the $[0, 1]$ range), producing two parallel scaled representations of each dataset that a downstream model could draw on.

9 Feature Selection

This introductory experiment focuses on exploring the Python data-science stack and identifying the correct ML task per dataset, so no feature-selection algorithm (e.g., `SelectKBest`, chi-square, ANOVA) is executed in the notebook itself. Based on the structure and correlation behaviour observed during EDA, the following techniques would be the natural next step for each dataset:

- **Loan Amount Prediction:** A correlation-based filter or `SelectKBest` with an F-regression score, since the target is continuous and `ApplicantIncome`, `LoanAmount` and `Credit_History` already show the clearest linear associations.
- **Predicting Diabetes:** `SelectKBest` with a chi-square or ANOVA F-test against the binary `Outcome` label, given that `Glucose` and `BMI` showed the strongest visual association with diabetes status.
- **Iris Dataset:** With only four true measurement features and no missing data, feature selection is largely unnecessary; if needed, ANOVA F-test ranks `PetalLengthCm` and `PetalWidthCm` as the most discriminative for species classification.

- **Email Spam Classification:** Since `v2` (the message text) is the only informative column, feature selection here really means text-vectorization (e.g., TF-IDF or bag-of-words) rather than filtering the current numeric columns, all of which carry little signal on their own.
- **Handwritten Character Recognition (MNIST):** Variance-threshold filtering is a natural first step, since a large number of border pixels (like the nine `1x1–1x9` columns examined in this notebook) are constant at 0 across every image and add no discriminative value; more advanced options include PCA for dimensionality reduction or CNN-based automatic feature learning.

10 Data Splitting

No explicit `train_test_split` call is present in this notebook, as the experiment stops at the EDA and preprocessing stage. For the supervised datasets in this batch (Loan Amount Prediction, Diabetes, Iris, Spam), an 80/20 or 70/30 train-test split with stratification on the target label (where the target is categorical) would be the standard next step, consistent with the split ratios used in the later experiments in this laboratory course. MNIST is conventionally distributed with a dedicated 60000-image training split (used here) and a separate 10000-image test split, so no additional splitting of the training file would ordinarily be required.

11 Model Identification

Table 5: Machine Learning Task Identification Summary

Dataset	Type of ML Task	Feature Selection Technique	Suitable ML Algorithm
Iris Dataset	Supervised – Multi-class Classification	ANOVA F-test / feature importance	K-Nearest Neighbors, Logistic Regression, SVM, Decision Tree
Loan Amount Prediction	Supervised – Regression	Correlation filter / SelectKBest (F-regression)	Linear Regression, Ridge/Lasso Regression, Random Forest Regressor
Predicting Diabetes	Supervised – Binary Classification	SelectKBest (Chi-square / ANOVA)	Logistic Regression, SVM, Random Forest Classifier
Classification of Email Spam	Supervised – Binary (Text) Classification	TF-IDF / Bag-of-Words vectorization	Naive Bayes, Logistic Regression, SVM
Handwritten Character Recognition / MNIST	Supervised – Multi-class (Image) Classification	Variance threshold / PCA	CNN, Random Forest, SVM, K-Nearest Neighbors

12 Performance Evaluation

Since no model was actually trained in this exploratory experiment, this section summarizes the metrics that would be appropriate once a model is fitted for each task, based on the task type identified above.

- **Regression (Loan Amount Prediction):** Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, and R^2 Score.
- **Binary Classification (Diabetes, Spam):** Accuracy, Precision, Recall, F1-score, and ROC-AUC – particularly important here since both datasets showed a moderate to significant class imbalance (34.9% positive for Diabetes, 13.4% spam for the SMS dataset).
- **Multi-class Classification (Iris, MNIST):** Accuracy, macro/weighted Precision, Recall and F1-score, and a Confusion Matrix to inspect per-class performance – especially useful for MNIST, where individual digit pairs (e.g., 4 and 9) are commonly confused.

13 Results

Table 6: Outlier Summary (1.5×IQR rule, selected columns)

Dataset	Columns with Notable Outlier Counts
Loan Amount Prediction	Education (134), Credit_History (89), Loan_Amount_Term (88), Gender (112), Self_Employed (82)
Predicting Diabetes	BloodPressure (45), Insulin (34), DiabetesPedigreeFunction (29), BMI (19)
Iris Dataset	SepalWidthCm (4); no outliers in any other column
Classification of Email Spam	v1 / ham-spam label (747); Unnamed: 2 (47)
Handwritten Character Recognition (MNIST)	None among the first ten inspected columns (label, 1x1–1x9), since the border pixels are constant

Across the five datasets, the EDA and preprocessing stage surfaced several consistent patterns. The Loan dataset combines multiple partially-missing categorical fields with a right-skewed income distribution. The Diabetes dataset contains a number of biologically implausible zero values (in Glucose, BloodPressure, SkinThickness, Insulin and BMI) that were treated as ordinary numeric values by the median-imputation logic in this notebook, since they are not stored as null. The Iris dataset is clean, small, and perfectly class-balanced. The Spam dataset is dominated by free-text content that the current numeric preprocessing pipeline does not meaningfully capture, and also contains 403 duplicate

rows that were identified but not removed in this pass. The MNIST dataset is very large and, for the specific subset of border pixels examined here, largely constant.

14 Discussion

The experiment involved analysing five different datasets with varying characteristics to understand how the machine learning workflow changes according to the type of data. The common steps of data loading, exploratory data analysis, visualization and preprocessing were applied to all datasets to study their structure and identify suitable machine learning approaches.

The Iris dataset was found to be a clean and balanced dataset with numerical features, making it suitable for classification algorithms with minimal preprocessing. The Loan Amount Prediction and Diabetes datasets required additional preprocessing steps such as handling missing values, encoding categorical attributes and scaling numerical features before applying machine learning models.

The Email Spam Classification dataset demonstrated that the choice of preprocessing technique depends on the nature of the data. Since it contains text information, techniques such as text feature extraction and vectorization are more appropriate compared to simple numerical encoding. The MNIST dataset highlighted the challenges of working with high-dimensional image data, where techniques like dimensionality reduction and specialized algorithms can improve efficiency.

The experiment also showed the importance of feature selection in reducing irrelevant information and improving model performance. By analysing dataset characteristics such as feature types, missing values, distributions and relationships between attributes, suitable machine learning tasks and algorithms were identified for each dataset.

Overall, the experiment provided a better understanding of how exploratory data analysis, preprocessing and feature selection form the foundation of an effective machine learning workflow.

15 Inference Table

Table 7: Inference Table – Dataset, ML Task and Recommended Approach

Dataset	Type of ML task	Feature Selection Technique	Suitable ML Algorithm
Iris Dataset	Classification (Multi-class, Supervised)	ANOVA F-test	K-Nearest Neighbors / SVM
Loan Amount Prediction	Regression (Supervised)	Correlation filter / SelectKBest	Linear / Ridge Regression
Predicting Diabetes	Classification (Binary, Supervised)	Chi-square / ANOVA SelectKBest	Logistic Regression / Random Forest
Classification of Email Spam	Classification (Binary, Text, Supervised)	TF-IDF vectorization	Naive Bayes / SVM
Handwritten Character Recognition / MNIST	Classification (Multi-class, Image, Supervised)	Variance threshold / PCA	CNN / Random Forest

This table directly mirrors the summary requested in the assignment brief and consolidates the task-identification exercise for all five datasets in a single, submission-ready reference.

16 Learning Outcomes

- Learned to use Python libraries such as NumPy, Pandas, SciPy, Scikit-learn and Matplotlib for machine learning tasks.
- Performed data loading, exploration and visualization using different datasets.
- Understood preprocessing techniques including handling missing values, encoding categorical variables and feature scaling.
- Identified suitable machine learning tasks and algorithms for different types of datasets.
- Learned the importance of feature selection and evaluation metrics before building machine learning models.
- Understood that different datasets require different preprocessing approaches depending on their data type and structure.

17 Conclusion

The experiment provided hands-on experience with essential Python libraries such as NumPy, Pandas, SciPy, Scikit-learn and Matplotlib for machine learning applications. Various datasets including Loan Amount Prediction, Diabetes Prediction, Iris, Email Spam Classification and MNIST were explored through data loading, exploratory data analysis, visualization and preprocessing techniques.

The appropriate machine learning task, feature selection approach and suitable algorithms were identified for each dataset based on their characteristics. This experiment emphasized the importance of data understanding, preprocessing and feature selection as essential steps before building effective machine learning models.

18 References

- [1] NumPy Developers, “NumPy Documentation.” Available: <https://numpy.org/doc/>
- [2] Pandas Development Team, “Pandas: 10 Minutes to pandas.” Available: https://pandas.pydata.org/docs/user_guide/10min.html
- [3] SciPy Community, “SciPy Reference Guide.” Available: <https://docs.scipy.org/doc/scipy/>
- [4] Scikit-learn Developers, “Scikit-learn: Machine Learning in Python.” Available: <https://scikit-learn.org/stable/>
- [5] Matplotlib Development Team, “Matplotlib Documentation.” Available: <https://matplotlib.org/stable/>
- [6] Kaggle, “Loan Prediction Problem Dataset.” Available: <https://www.kaggle.com/>
- [7] UCI Machine Learning Repository, “Pima Indians Diabetes Dataset.”
- [8] UCI Machine Learning Repository, “Iris Dataset.”
- [9] Kaggle, “SMS Spam Collection Dataset.”
- [10] Y. LeCun, C. Cortes, and C. Burges, “MNIST Handwritten Digit Database,” Available: <http://yann.lecun.com/exdb/mnist/>